STIzen: AN INTEGRATED WEB-BASED ACADEMIC SCHEDULING MANAGEMENT SYSTEM FOR STI COLLEGE NAGA

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Presented to the Faculty of
Information Technology Department
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of the Requirements of the Degree
Bachelor of Science in Information Technology

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CHAPTER I

INTRODUCTION

In today's digital age, educational institutions increasingly rely on technology to improve efficiency, accuracy, and overall management of academic processes. One of the most critical functions in any academic institution is scheduling, which involves organizing classes, faculty assignments, classrooms, and examinations. An effective scheduling system plays a vital role in ensuring smooth academic operations, preventing conflicts, and maintaining a balanced workload among faculty members.

However, managing academic schedules can be a complex and time-consuming task, especially when handled through manual or traditional methods such as spreadsheets. Without a centralized system, institutions may encounter issues such as overlapping schedules, room conflicts, delayed updates, and difficulty in coordinating across departments. These challenges highlight the importance of having a centralized scheduling system that can streamline processes and improve communication among stakeholders.

In response to these needs, this study introduces a web-based academic scheduling system designed to enhance efficiency, accuracy, and coordination. By integrating automation and validation mechanisms, the system aims to reduce scheduling conflicts, minimize administrative workload, and provide a more reliable solution for academic institutions. In addition, the system features a fully responsive web interface that ensures all users can conveniently access their schedules, notifications, and announcements using any device, including desktops, tablets, and mobile phones.

Background of the Study

Academic scheduling is an essential administrative function that focuses on organizing institutional resources, including classrooms, faculty, and students. It involves arranging class timetables, aligning program requirements, and distributing teaching loads effectively. However, many institutions still depend on manual processes or basic spreadsheet tools, which can lead to delays, inconsistencies, and frequent adjustments. Despite the availability of modern educational technologies, the use of fully automated or intelligent web-based scheduling systems remains limited in many local schools and colleges.

At STI College Naga, the scheduling process is largely manual and handled by academic heads and department heads, with support from registrars during enrollment adjustments. This process becomes increasingly difficult when managing multiple academic programs, including Senior High School strands such as Accountancy, Business, and Management (ABM), Science, Technology, Engineering, and Mathematics (STEM), Humanities and Social Sciences (HUMSS), and Mobile Application and Web Development (MAWD), and college programs such as Bachelor of Science in Information Technology (BSIT) and Bachelor of Science in Hospitality Management (BSHM). The absence of an integrated system makes it challenging to coordinate schedules across different departments, resulting in issues such as overlapping classes, room conflicts, and unbalanced faculty workloads. Survey results indicate that resolving these conflicts manually can take hours or even an entire day, which demonstrates the inefficiency of the current approach. This highlights the need for a more efficient and centralized system that can streamline scheduling processes.

In addition to scheduling complexity, communication and monitoring also present significant challenges. Faculty members and students often experience confusion due to sudden schedule changes, unclear room assignments, and delays in receiving updates. Without a centralized platform, announcements and schedule changes are not efficiently disseminated, leading to inconsistencies in information. Furthermore, administrators lack real-time visibility of scheduling activities, such as room utilization and faculty loading, making it difficult to make timely and informed decisions. The absence of system-generated notifications and a unified dashboard further limits coordination among stakeholders. Students, in particular, experience inconvenience in accessing timely updates when relying only on non-responsive web platforms that are difficult to navigate on mobile devices. These gaps underscore the urgent need for improved communication and centralized access to academic information. This situation emphasizes the importance of providing real-time and accessible updates to ensure effective coordination among all users.

Another critical issue is the lack of transparency and accountability in the current process. Since scheduling changes are handled manually, there is limited tracking of modifications, making it difficult to identify errors or review previous decisions. The absence of audit mechanisms and version control results in reduced traceability and increased risk of inconsistencies in scheduling data. Additionally, managing faculty workloads, room availability, and subject requirements without system support makes it difficult to ensure fairness, accuracy, and compliance with institutional policies. To address the identified challenges, there is a clear need for a centralized and semi-automated web-based scheduling system that integrates key functionalities to support academic operations. Such a system is expected to streamline scheduling processes by reducing manual intervention and ensuring that all academic resources such as faculty, classrooms, and schedules are efficiently managed. It should incorporate features like automated draft schedule generation, preventive conflict validation, intelligent time-slot and resource suggestions, faculty load monitoring, and real-time notifications. It should also provide tools for managing examination schedules, rooms, subjects, and sections, while offering a unified dashboard for monitoring institutional data. Moreover, incorporating audit logs, role-based access control, and secure data handling can enhance system transparency, accountability, and security. By providing a single platform that combines automation with user control, such a system can significantly reduce errors, save administrative time, and improve coordination across all academic departments.

Based on the identified challenges, STIzen is proposed as a comprehensive web-based academic scheduling system tailored for STI College Naga. The system is designed not only to automate scheduling processes but also to enhance communication and accessibility among stakeholders within the institution. Through features such as real-time notifications, centralized announcements, PDF report generation, and data archiving, STIzen supports both day-to-day scheduling operations and long-term academic management. The system's fully responsive design ensures that all users can conveniently access their schedules, receive real-time notifications, and stay updated on announcements using any device, including desktops, tablets, and mobile phones, anytime and anywhere. By addressing the limitations of existing manual processes, the system aims to provide a scalable, reliable, and user-centered solution that enhances the overall scheduling experience for administrators, faculty, and students.

Research Questions

This study aims to develop STIzen: An Integrated Web-Based Academic Scheduling Management System for STI College Naga to improve the current manual scheduling process, minimize conflicts, and efficiently manage academic resources. Specifically, it seeks to answer the following questions:

1. How can a web-based platform centralize and efficiently manage class and examination schedules, including preliminary, midterm, pre-final, final, and special examinations for both Senior High School and College programs?
2. How can preventive validation mechanisms and intelligent time-slot suggestions reduce scheduling conflicts involving faculty, rooms, sections, and time allocation?
3. How can the system effectively monitor and manage faculty workloads, including teaching assignments, lecture and laboratory units, and overload conditions?
4. How can role-based access control (RBAC) be implemented to regulate user permissions for the School Administrator, Academic Head, Department Heads, Registrar, Faculty, and Students to ensure secure and controlled access to scheduling functions?
5. How can the system provide real-time notifications and accessible schedule information, including viewing and PDF export features, for faculty and students, ensuring that users are consistently informed of any updates, changes, or assigned schedules across different devices and platforms?
6. How can audit logs and archiving features ensure transparency, accountability, and proper tracking of all scheduling activities?

Significance of the Study

The study on STIzen is significant because it directly benefits various stakeholders within STI College Naga by improving the efficiency, clarity, and reliability of academic scheduling. It highlights the importance of an organized and systematic approach in managing academic schedules within the institution.

School Administrator. The School Administrator gains a centralized platform to view schedules, ensuring all academic planning complies with institutional policies. The system provides real-time visibility of schedule drafts, notifications, and audit logs, which support timely decision-making, accountability, and transparency.

Academic Head. As the primary scheduler, the Academic Head benefits from automated conflict prevention, workload computation, and intelligent scheduling suggestions through the auto-suggestion engine. STIzen streamlines the creation and management of schedules across Senior High School strands and College programs, reducing manual errors and allowing more focus on strategic academic planning. Real-time notifications further assist the Academic Head in monitoring schedule updates and changes submitted by Department Heads and the Registrar.

Registrar. The Registrar is supported through accurate tracking of subjects, sections, and room assignments. The system reduces time spent on manual adjustments, prevents overlapping schedules, and ensures proper enrollment management, helping maintain data accuracy and administrative efficiency. Notifications provide timely updates regarding schedule publication and modifications.

Department Head. Department Heads can draft and propose schedules for their respective programs while collaborating with the Academic Head. The system ensures that their contributions are integrated without causing conflicts and that finalized schedules remain secure until approved, thanks to role-based access control and locked schedule functionality. They also receive notifications regarding approval status, updates, and finalized schedules.

Faculty. Faculty members benefit from automated workload monitoring, teaching load summaries, and digital acknowledgment of assigned duties. These features prevent overload, ensure fair distribution of responsibilities, and allow faculty to verify their schedules quickly and conveniently. Role-based notifications inform faculty of their assigned teaching schedules, proctoring duties, and any schedule updates.

Students. Students gain easy access to their class and examination schedules through the web portal, which is fully responsive and accessible on any device including mobile phones and tablets. Real-time notifications and PDF export options provide clarity, reduce confusion, and support better time management.

STI College Naga. The college benefits from a more organized and transparent scheduling process. Features such as preventive conflict management, optimized room and faculty utilization, centralized announcements, and audit trails contribute to a smoother academic workflow, improving overall operational efficiency. The inclusion of real-time notifications enhances communication across all stakeholders. Additionally, the system supports better planning and decision-making by providing administrators with timely and reliable scheduling information.

Future Researchers. This study serves as a valuable baseline for future developments in academic management systems. It provides a blueprint for integrating multi-departmental requirements (SHS and College) and tiered administrative roles, which other researchers can build upon for additional features such as attendance tracking, payroll integration, or AI-driven predictive scheduling.

Objectives of the Study

The main objective of this study is to develop STIzen: An Integrated Web-Based Academic Scheduling Management System for STI College Naga that addresses the challenges of manual scheduling, scheduling conflicts, faculty load management, and accessibility of academic information. Specifically, the study aims to achieve the following:

1. To design and implement a centralized web-based academic scheduling system that integrates Senior High School and College programs into a single platform, allowing efficient access and management of class and examination schedules.
2. To develop intelligent scheduling and conflict prevention mechanisms that detect and avoid overlaps in faculty assignments, room usage, section schedules, and time slots, while providing optimized scheduling suggestions.
3. To implement a faculty workload management feature that computes and monitors teaching loads, lecture and laboratory units, proctoring duties, and overload conditions in real time to ensure balanced workload distribution.
4. To establish a role-based access control (RBAC) system that defines and enforces permissions for School Administrator, Academic Head, Department Heads,

Registrar, Faculty, and Students, ensuring secure and controlled access to scheduling functions.

5. To develop a responsive web-based portal that provides real-time access to class and examination schedules, delivers system notifications for schedule updates and assignments, and allows users to export schedules in PDF format for offline access across desktop, tablet, and mobile devices.
6. To ensure system security, data integrity, and accountability through audit logs, schedule versioning, and academic year archiving, preserving historical records while maintaining system reliability and performance.

Scope and Limitations

The study focuses on the development of STIzen: An Integrated Web-Based Academic Scheduling Management System for STI College Naga to improve the efficiency, accuracy, and accessibility of academic scheduling. The system provides a centralized platform for managing class and examination schedules for both Senior High School and College programs, supporting schedule creation, editing, locking, and publishing in an organized and systematic manner.

It incorporates conflict detection and time-slot suggestion features to prevent scheduling overlaps involving faculty, rooms, sections, and time allocation, helping improve the overall scheduling process. The system also supports examination scheduling for prelim, midterm, pre-final, final, and special examinations, including room assignments and faculty proctor management to ensure proper coordination.

In addition, it includes modules for faculty management, room management, course and section management, and subject management to ensure proper allocation and monitoring of academic resources across departments. Role-Based Access Control (RBAC), audit logs, and academic year archiving are implemented to ensure security, accountability, and historical data tracking for better system governance and record-keeping.

The system is developed as a responsive web-based platform, allowing users to conveniently access schedules and related information through desktop, tablet, and mobile devices, ensuring flexibility and accessibility across different environments. The system also supports PDF export functionality, enabling faculty and students to download and print their class and examination schedules for offline reference.

Despite its features, the system has several limitations. The automated scheduling and conflict detection mechanisms are rule-based and may not fully address complex or unexpected situations such as sudden enrollment changes, faculty unavailability, or limited room resources. Manual intervention from authorized users is still required for special cases, which may affect the speed of schedule updates.

The system also depends on stable internet connectivity, and interruptions may impact access to schedules and notifications. Furthermore, data accuracy relies on user input, and errors in encoding faculty, subjects, or sections may still affect scheduling results. The system is specifically designed for STI College Naga and may require modification for significant institutional changes or expansion. Lastly, while the system provides scheduling support, final decisions still rely on human judgment, particularly in irregular or cross-department scheduling scenarios.

Definition of Terms

To ensure a clear and consistent understanding of the technical and administrative concepts presented in this study, the following terms are defined according to their specific application within the STIzen system:

Archiving – The process of storing historical schedules from past academic years in the system while maintaining performance for current scheduling operations. This ensures that old data can be accessed for reference or reporting without affecting active schedules.

Audit Trail – A chronological record of all system activities, capturing the user, action performed, and timestamp for each event, ensuring transparency and accountability. This feature helps administrators track changes in the system, detect unusual activities, and support troubleshooting when issues arise.

Auto-Suggestion Engine – A role-based feature that automatically recommends optimal time slots, classrooms, and faculty assignments to help administrators create efficient schedules while following institutional rules.

Notifications – Automated alerts sent to users based on their role, informing them of schedule updates, published schedules, assigned classes, or proctoring duties. This ensures timely communication and reduces confusion.

Password Hashing – Security method that stores encrypted versions of user passwords. It ensures that even if the database is compromised, the original passwords cannot be easily retrieved. This adds an extra layer of protection by requiring attackers to decode the hash, which is computationally difficult and time-consuming.

Preventive Conflict Management / Conflict Detection – A system function that automatically blocks scheduling conflicts in faculty assignments, classroom allocations, and section timetables at the point of entry, preventing errors before they occur. This helps ensure reliable and conflict-free schedule generation throughout the system.

Role-Based Access Control (RBAC) – A security mechanism that allows only authorized users to perform specific actions in the system according to their assigned roles. This ensures proper control, prevents unauthorized changes, and maintains the integrity of the system's data.

Schedule Versioning – A system function that maintains previous versions of schedules, allowing tracking of changes over time and recovery of past schedules if necessary. This helps in auditing and restoring schedules after modifications.

Single Sign-On (SSO) – An authentication method that allows users to access the STIzen system using their institutional STI email and password. It supports two domains: @naga.sti.edu for administrative users (School Administrator, Academic Head, Registrar, Department Head) and @naga.sti.edu.ph for faculty and students.

System Administrator (Academic Head) – Refers to the Academic Head who serves as the primary system administrator of STIzen. This role is responsible for managing user accounts, overseeing scheduling operations, and maintaining system control and access permissions.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a structured and critical review of related literature and studies that serve as the foundation for the proposed system, STIzen. The review is organized into three main sections: local studies, foreign studies, and related systems and technologies. It concludes with a synthesis that highlights the research gaps and justifies the development of the proposed system.

The primary objective of this chapter is not only to present previous works but also to analyze their methodologies, strengths, and limitations. This analysis provides a clear understanding of how existing systems address academic management challenges and identifies areas where improvements are necessary. These insights guide the development of STIzen as an integrated and efficient academic management solution.

Local Studies

The development of academic scheduling systems in the Philippines has increasingly focused on digitalizing administrative workflows to address the complexities of local institutional requirements. Recent local literature emphasizes the transition from manual, paper-based processes to web-based and cloud-integrated platforms aimed at improving resource coordination and data security. While these studies demonstrate significant progress in centralizing records and automating basic scheduling tasks, they also reveal persistent challenges in real-time conflict prevention, system transparency, and user accessibility. The following studies provide a comprehensive overview of the current technological landscape within the Philippine academic setting and identify the functional gaps that modern scheduling solutions must address.

To begin with, Garcia et al. (2022) developed a web-based academic scheduling system that centralized class assignments, faculty load, and room allocation using a database-driven structure. The system enforced constraint validation to minimize overlapping schedules, resulting in improved accuracy and reduced manual errors. However, conflict detection was applied only after schedules were generated, limiting its effectiveness in preventing inconsistencies during the encoding process. This limitation emphasizes the need for preventive conflict detection and real-time validation, supported by intelligent suggestions to guide users in creating accurate and conflict-free schedules.

Similarly, Reyes (2021) designed an integrated academic management system that incorporated role-based access control to regulate user permissions across administrative functions. The system improved data security and coordination by ensuring that only authorized users could access and modify records. Despite these strengths, it lacked audit tracking and version monitoring, making it difficult to trace changes and review scheduling history. This highlights the importance of integrating audit trails and version control mechanisms to ensure accountability, transparency, and secure management of scheduling data.

Moreover, Santos (2021) introduced a web-based scheduling platform that enabled users to manage academic-related schedules through a centralized interface. The system improved accessibility and reduced scheduling conflicts by organizing data within a single repository. However, it did not include intelligent recommendation features to assist users in selecting optimal schedules based on constraints. This limitation underscores the need for auto-suggestion mechanisms that can support decision-making and enhance scheduling efficiency.

In addition, Dela Cruz and Mendoza (2023) developed a cloud-based academic information system that integrated scheduling, reporting, and user authentication modules. The system enabled real-time access to institutional data, improving communication between administrators and faculty members. Nevertheless, it lacked audit logging and version tracking capabilities, making it difficult to monitor system changes and maintain historical records. This limitation highlights the importance of incorporating audit trails and version control to ensure data integrity and traceability in scheduling systems.

Furthermore, Torres et al. (2024) implemented a multi-user academic system that incorporated real-time notification features to inform users of schedule updates and changes. The system enhanced coordination and reduced miscommunication by delivering timely updates to stakeholders. However, it did not include preventive conflict detection, as scheduling inconsistencies were identified only after they occurred. This emphasizes the need for systems that combine real-time notifications with proactive validation to prevent conflicts before they affect operations.

Likewise, Bautista and Ramos (2022) explored the use of role-based access control combined with audit logging in institutional systems to improve data security and accountability. The system effectively tracked user actions, ensuring transparency and proper monitoring of system activities. However, the complexity of role configuration posed usability challenges, particularly for non-technical users. This highlights the need for simplified RBAC implementation integrated with user-friendly interfaces to maintain both security and accessibility.

Foreign Studies

Globally, the academic community has made significant strides in addressing the complexities of timetabling through advanced computational models and automated frameworks. International research often shifts beyond simple digitalization, focusing instead on optimization algorithms, mathematical programming, and multi-objective scheduling to handle the vast constraints of large-scale institutions. These foreign studies provide a standard for high-level accuracy and resource optimization; however, they frequently highlight a disconnect between algorithmic power and practical administrative usability. By reviewing these global advancements, the specific technical requirements for creating a responsive, interactive, and reliable scheduling system become more evident. The following studies examine various international approaches to exam and course scheduling, highlighting the necessary integration of proactive validation and real-time stakeholder communication.

Mohammed et al. (2022) developed an examination scheduling system that organized sessions based on institutional constraints such as room capacity and time availability. The system produced feasible schedules that successfully improved administrative efficiency in managing complex exam periods. However, it lacked a real-time validation layer, which meant conflicts were identified only after the generation process was complete. This limitation justifies the implementation of STIzen's preventive validation, which blocks scheduling inconsistencies at the point of data entry to ensure immediate accuracy.

Similarly, Siew et al. (2024) analyzed various scheduling methodologies, concluding that effective systems must balance mathematical precision with practical usability. Their research revealed that many high-level optimization tools are difficult for non-technical staff to navigate due to a lack of interactive support. This underscores the importance of the Auto-suggestion module in STIzen, which provides users with proactive guidance to simplify decision-making. By offering intelligent recommendations, the system reduces the cognitive load on administrators while maintaining high scheduling standards.

Moreover, Steiner et al. (2024) introduced a curriculum-based model designed to prevent student schedule overlaps by analyzing course combinations. While the model achieved high feasibility across complex academic structures, it required intricate manual configurations that hindered rapid adjustments. This highlights the need for STIzen's intelligent suggestion features, which allow users to manage complex constraints through a simplified interface. Integrating these features ensures that institutional rules are followed without sacrificing the flexibility needed for mid-semester changes.

Furthermore, Davison et al. (2025) explored a scheduling framework capable of managing both physical and online classes within a unified environment. The study found that hybrid learning models significantly increased system complexity, requiring advanced coordination among multiple user groups. This emphasizes the necessity of STIzen's Real-time Notifications, which ensure that updates are instantly synchronized across all student and faculty portals. Prompt communication remains the primary defense against the information silos that often occur in complex scheduling environments.

Additionally, recent research by Gu et al. (2025) highlights the integration of security layers and monitoring mechanisms as essential components of modern scheduling platforms. While these advancements improve data integrity, the study noted that many systems still fail to provide a comprehensive history of administrative actions. This gap validates the inclusion of Audit Trails and Version Control in STIzen, ensuring every modification is traceable and reversible. Maintaining a transparent record of system changes is critical for both institutional accountability and data recovery.

Related Systems

The global market for academic and resource management software has established a high standard for automation, security, and real-time data synchronization. By analyzing industry-leading platforms and established technological frameworks, the functional requirements for a comprehensive scheduling solution become more clearly defined. These related systems demonstrate how features such as conflict validation engines, role-based security, and automated communication hubs are implemented in commercial and open-source environments. Reviewing these existing solutions provides a benchmark for the development of STIzen, ensuring that its architecture aligns with modern administrative needs while addressing the specific usability gaps found in current market offerings. The following systems represent the state-of-the-art in institutional management and provide the technical rationale for the integrated features of the proposed system.

To begin with, TimeEdit (2025) is a global scheduling platform that manages class and examination timetables through configurable constraints such as room capacity and instructor availability. The system identifies conflicts before schedules are finalized, significantly improving accuracy and consistency in timetable creation. It also provides centralized access, allowing users to view updates in real time across the institution to reduce communication gaps. However, the system requires extensive manual configuration, highlighting the importance of STIzen’s flexible scheduling controls combined with automated preventive conflict detection.

Similarly, Celcat Timetabler (2024) supports academic scheduling by analyzing complex relationships among rooms, faculty, and courses within a unified database. It enables the early identification of overlaps and assists users in selecting appropriate scheduling options through a centralized interface. The platform also implements controlled multi-user access to maintain data integrity across various academic departments. Despite these strengths, its high learning curve highlights the need for STIzen's simplified interface combined with auto-suggestion and RBAC mechanisms to support efficient system use by all staff members.

Moreover, Coursedog (2025) provides a centralized platform that integrates scheduling, workflow management, and real-time updates for higher education institutions. The system maintains a detailed history of scheduling changes, allowing administrators to track modifications and review previous iterations of a timetable. This improves institutional transparency and supports better data-driven decision-making when managing complex schedules. However, its heavy reliance on cloud-sync configuration underscores the importance of STIzen’s built-in audit trail and version control features to ensure consistent and reliable local system tracking.

In addition, Microsoft Azure Active Directory (2024) demonstrates the industry-standard implementation of SSO and RBAC for secure system entry. The platform allows users to access multiple services using a single set of credentials while strictly enforcing permission levels based on assigned roles. This improves both security and user convenience by reducing the need for multiple passwords and minimizing the risk of unauthorized data access. Nevertheless, the complexity of its backend highlights the need for STIzen's well-structured, pre-configured RBAC and SSO integration to maintain both high usability and robust data protection.

Furthermore, Firebase Cloud Messaging (2024) enables real-time communication by delivering instant push notifications to users regarding critical system updates and schedule changes. The system ensures the timely dissemination of information, which improves institutional coordination and reduces delays in faculty-student communication. It supports scalable notification delivery across multiple platforms, ensuring that every stakeholder remains informed regardless of their device. However, the potential for notification fatigue emphasizes the importance of STIzen's controlled notification system, which balances immediate responsiveness with a clean user experience.

Finally, the Ellucian Banner Student Self-Service Portal (2025) provides a good example of device-responsive accessibility. The system is designed to automatically adjust complex academic schedules, so they are easy to read on any screen size. This is important for the present study because it shows how large data, such as weekly class schedules, can still be viewed clearly on small devices like smartphones. Therefore, by applying this kind of responsive design, the STIzen system can ensure that both faculty using laptops and students using mobile phones can easily access and understand their schedules.

Synthesis

The reviewed literature and studies reveal a critical shift in academic management, moving from manual, error-prone scheduling toward integrated, secure, and automated digital ecosystems. Both local and international research emphasize that as educational institutions expand—incorporating hybrid learning models and complex resource constraints—traditional "reactive" methods of scheduling are no longer viable for maintaining operational efficiency.

A primary theme across local studies, such as Garcia et al. (2022) and Torres et al. (2024), is the urgent need for Preventive Conflict Detection. While current local systems have successfully centralized institutional data, they frequently identify overlaps only after a schedule is generated, leading to repetitive manual corrections. This functional gap justifies the proactive validation layer in STIzen, which blocks inconsistencies at the point of data entry. Furthermore, the findings of Santos (2021) and Siew et al. (2024) highlight a lack of decision-support tools, reinforcing the necessity for Auto-suggestion features to assist administrators in optimizing faculty and room assignments.

Global research and related industrial systems provide the technical benchmarks for security, accessibility, and synchronization. The research by Mohammed et al. (2022) and the implementation of Microsoft Azure AD (2024) confirm that RBAC and SSO are indispensable for protecting data integrity in multi-user environments. Additionally, the increasing complexity of hybrid academic structures explored by Davison et al. (2025) underscores the vital role of Real-time Notifications. The success of platforms like Firebase Cloud Messaging (2024) demonstrates how instant synchronization can bridge the communication gap between administrators and stakeholders when schedules are modified.

Finally, the literature identifies a critical requirement for institutional accountability, accessibility, and data resilience. Industry systems such as Coursedog (2025) emphasize the importance of tracking administrative actions to ensure transparency and error recovery. In addition, the Ellucian Banner Student Self-Service Portal (2025) provides a strong example of device-responsive accessibility, where complex academic schedules are automatically adjusted for readability across different screen sizes. This demonstrates that even large datasets, such as weekly class schedules, can remain clear and user-friendly on mobile devices. The absence of these features in many local implementations, as noted by Dela Cruz and Mendoza (2023) and Reyes (2021), highlights the importance of STIzen's Audit Trail, Version Control, and Responsive Design modules. By integrating these features, the proposed system creates a secure, traceable, and accessible environment that empowers administrators and users to manage academic operations with higher precision and significantly reduced operational risk.

Research Gap

While existing local and international studies have successfully transitioned academic scheduling from manual to digital platforms, several key gaps remain. Local studies such as Garcia et al. (2022) and Torres et al. (2024) show a post-generation conflict detection approach, where errors are only identified after schedules are saved, resulting in inefficient revisions. Meanwhile, international systems like Steiner et al. (2024) provide high accuracy but present a usability gap due to complex configurations that are difficult for non-technical users.

Additionally, many systems lack integrated accountability, accessibility, and real-time decision support. As noted by Dela Cruz and Mendoza (2023), most local frameworks still have weak Audit Trails and Version Control, limiting transparency and error recovery. Furthermore, several systems do not support device-responsive design, making schedules less accessible on mobile devices. Although platforms like TimeEdit (2025) and Coursedog

(2025) address some of these features, their high cost and reliance on external infrastructure reduce their suitability for local institutions.

STIzen addresses these gaps by implementing Preventive Conflict Detection and Auto-suggestions within a user-friendly, responsive interface, ensuring conflict-free and accessible scheduling at the point of data entry. Combined with Audit Logging and Role-Based Access Control, the system offers a secure, transparent, and practical solution for academic scheduling.

Conceptual Framework

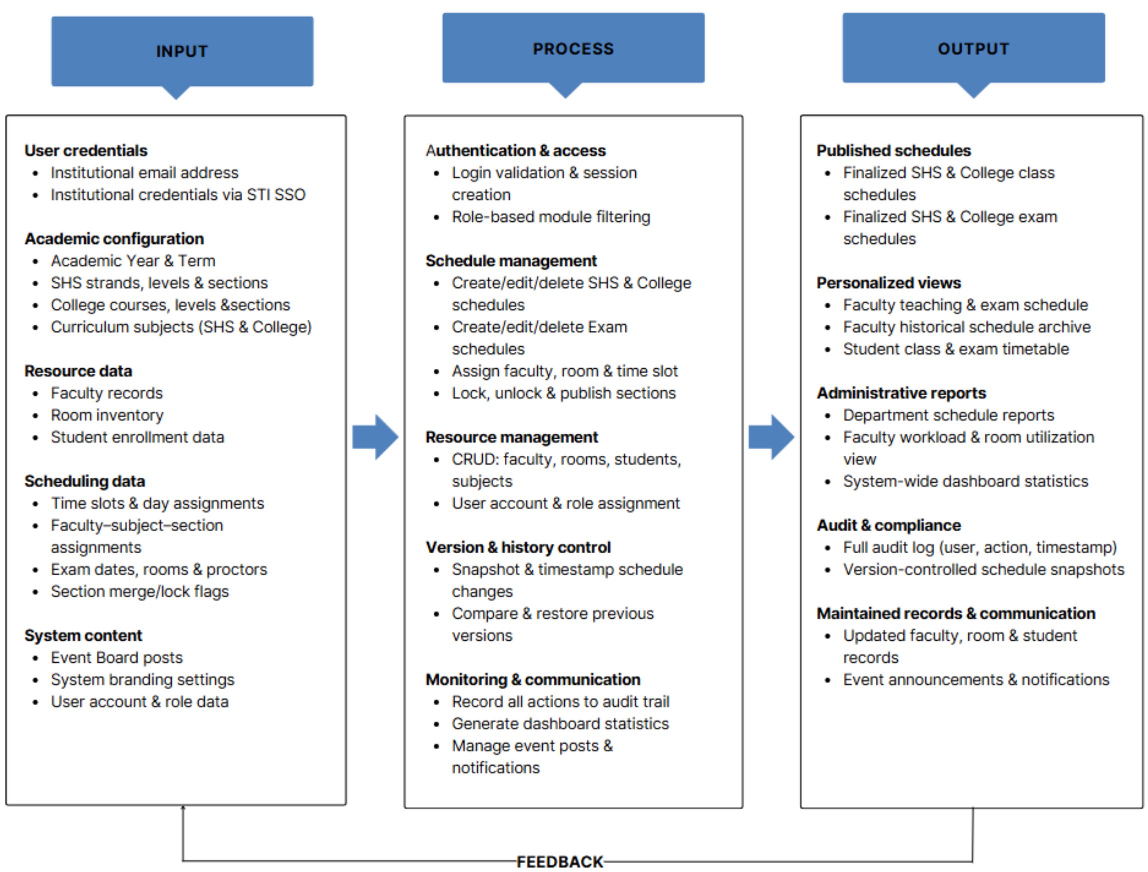

Figure 1. The STIzen Conceptual Framework: IPO Model

Figure 1 above shows the conceptual framework of STIzen using the Input-Process-Output (IPO) model, which illustrates how the system receives data, processes it, and produces meaningful outputs. This model maps the flow of information from initial academic and resource data input to the generation of finalized schedules and administrative reports. It provides a structured view of how the system supports the scheduling needs of both Senior High School and College programs at STI College Naga.

In the first phase, the Input component encompasses all data and actors that enter the system, including user credentials authenticated via STI SSO, academic configuration data such as academic year, term, strands, courses, and sections, resource data covering faculty records, room inventory, and student enrollment, scheduling data including time slots, faculty-subject-section assignments, exam dates, and section lock flags, as well as system content such as event board posts, branding settings, and user account data. These inputs collectively represent the foundational data requirements of the system, ensuring that all scheduling decisions are based on accurate and up-to-date information. The quality and completeness of the input data directly influence the reliability and effectiveness of the outputs generated by the system.

Proceeding to the next phase, the Process component represents the core operations performed by the system upon the data it receives. These include login validation and role-based access control, creation and management of SHS, College, and Exam schedules, CRUD operations on all academic resources, version control through schedule snapshots and restoration, and monitoring functions such as audit logging, dashboard statistics generation, and event notification management. Each process is governed by a defined set of role-based permissions, ensuring that only authorized users can perform specific actions within the system, thereby maintaining data security and integrity. The processing layer acts as the engine of the system, where raw inputs are validated, organized, and transformed into structured schedule data that can be published and accessed by the appropriate user roles. The processing layer also supports a fully responsive front-end interface, ensuring that schedules, notifications, and announcements are properly displayed and accessible across all screen sizes and devices.

In the final phase, the Output component reflects the results and value delivered to each user role upon the completion of system processes. These include finalized and published class and exam schedules, personalized timetable views for faculty and students, department and administrative reports, a full audit trail with version-controlled schedule snapshots, updated academic records, and school-wide event announcements and notifications. These outputs are accessible through the fully responsive web-based platform, ensuring that students can conveniently access their schedules, notifications, and announcements on any device, including desktops, tablets, and mobile phones.

Finally, the feedback loop, shown as the arrow connecting Output back to Input, represents the system's iterative process where outputs such as user feedback, evaluation results, and audit reports are analyzed and used as new inputs. This allows STIzen to remain adaptive and continuously improve scheduling rules, resource management, and user experience based on real-world performance and user insights. It ensures that the system evolves through continuous refinement based on actual institutional needs and user behavior, supporting long-term system improvement and stability.

CHAPTER III

METHODS

This chapter presents the research methodology used in the development of the STIzen system. It explains the research design, data collection procedures, system development approach, tools used, and evaluation methods applied to ensure that the system effectively addresses the identified problems in academic scheduling at STI College Naga.

Development Methodology

In developing the STIzen system, the researchers will adopt the Modified Waterfall Model. This model is a linear and sequential software development methodology that will allow flexibility by permitting phases to overlap and permits revisiting earlier stages for verification, validation, and refinements. The approach ensures that the system is systematically developed while incorporating user feedback and maintaining alignment with institutional requirements.

The Modified Waterfall Model is particularly suitable for STIzen because it will provide a structured framework that supports careful planning, design, and testing of complex academic scheduling functionalities. By following this methodology, the researchers can clearly define each phase—from requirements gathering to deployment and maintenance—while ensuring that critical features like conflict detection, smart scheduling suggestions, role-based access, real-time notifications, and audit trails are implemented correctly. This approach will also allow iterative verification, so adjustments can be made early, reducing the risk of costly errors and ensuring the system meets the practical needs of administrators, faculty, and students at STI College Naga.

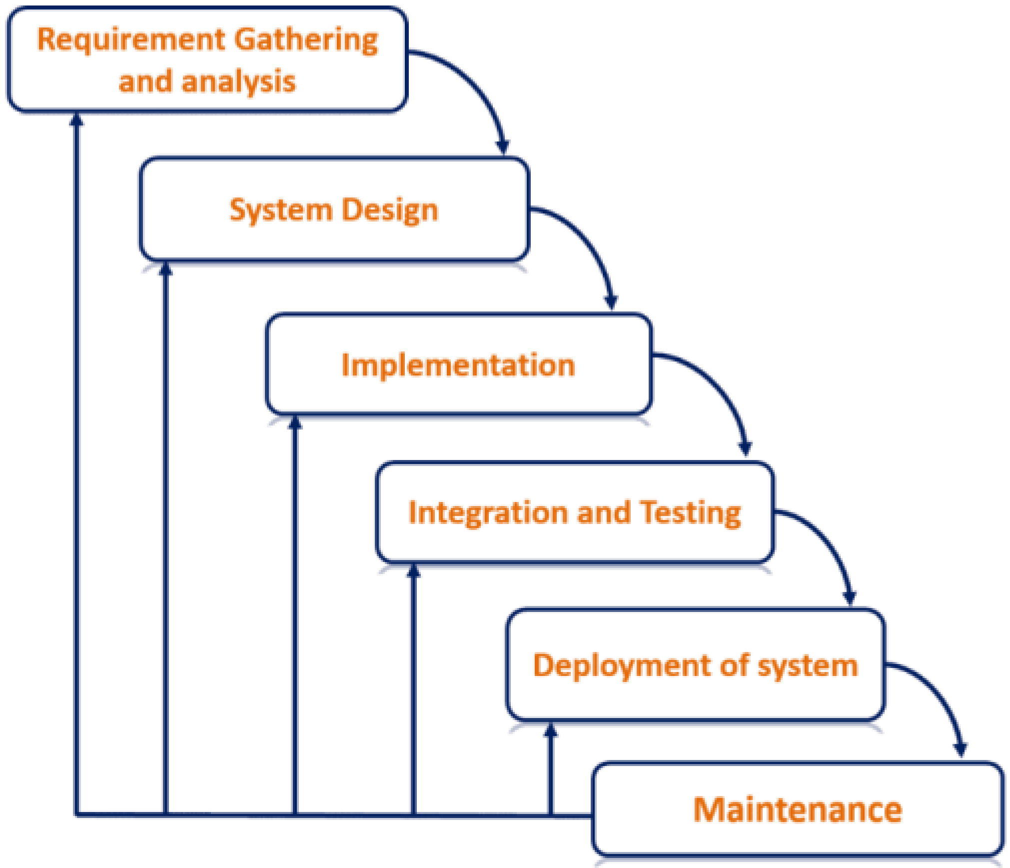

Figure 2. Modified Waterfall Model

Figure 2 above illustrates the Modified Waterfall Model used in the development of STIzen, highlighting a structured and sequential approach to software development. This model ensures that each phase builds upon the previous one while maintaining continuous refinement and alignment with stakeholder needs.

The first phase, Requirements Gathering and Analysis, will involve extensive consultation with key stakeholders, including administrators, faculty, and students, through interviews, surveys, observations, and document reviews to identify issues in the current manual scheduling process such as overlapping classes, faculty overload, double-booked rooms, and inefficient communication. Stakeholder input will define functional requirements including role-based access control, schedule versioning, notifications per user role, and faculty workload computation, while usability needs such as intuitive navigation, responsive design, and personalized dashboards will ensure high adoption. Technical constraints like secure STI Office 365 (O365) login, password hashing, and system compatibility will also be considered to ensure proper system integration.

In the System Design phase, these requirements will be translated into a detailed system architecture. This will include database design to manage programs, subjects, sections, classrooms, faculty, and schedules, with relationships to track room availability, faculty assignments, and historical records. Core functional modules will be specified, such as automated conflict detection, smart scheduling engines, faculty workload monitoring, notifications, and audit trail management.

Security and workflow considerations will be integrated to enforce role-based access and version control for schedule revisions. The design phase will produce a structured blueprint, ensuring scalability, maintainability, and alignment with stakeholder expectations. This architecture also incorporates responsive design principles to ensure the system's interface adapts seamlessly across different screen sizes and devices, providing a consistent user experience for all stakeholders.

During the Implementation phase, the system design will be converted into a functional web-based platform. Core modules will be coded, responsive student and faculty portals will be developed, and preventive mechanisms will be implemented to block overlapping classes or double-booked rooms. Existing scheduling data from Excel and departmental records will be migrated to maintain continuity.

The Integration and Testing phase will ensure that all components work seamlessly and meet performance expectations. Unit testing will validate individual modules like conflict detection, notifications, and workload computations. Integration testing will verify communication between modules, while system testing will assess overall performance under real-world scheduling scenarios. User Acceptance Testing (UAT) will involve administrators, department heads, faculty, and students to confirm usability, accuracy, and satisfaction of the web-based system across different devices and screen sizes.

In the Deployment phase, STIzen will be launched in a controlled environment at STI College Naga. A pilot rollout will allow monitoring of system performance and usability, while training sessions will familiarize stakeholders with features such as real-time notifications, PDF exports, and workload dashboards. Continuous monitoring during the first live scheduling cycles will help identify and resolve operational bottlenecks.

Finally, the Maintenance phase will ensure the long-term reliability and usability of STIzen through continuous support, including bug fixes, performance optimization, and feature enhancements. User feedback will be regularly collected to improve the system and adapt to evolving needs. Audit trails, data backups, version control, and security measures will maintain data integrity, while periodic updates will keep the system aligned with technological advancements and institutional requirements. This phase also ensures that the web-based system remains fully functional, secure, and responsive across all devices and user demands over time.

Software Requirements

The development and implementation of STIzen rely on a combination of software tools and technologies that support front-end and back-end development, database management, testing, collaboration, system design, and PDF generation. These tools collectively ensure the system is functional, secure, responsive, and user-friendly, ensuring efficient development and seamless system integration across all components.

Category	Software/Tools
Integrated Development Environment (IDE)	Visual Studio Code
Front-End Development	HTML5, CSS3, JavaScript, Bootstrap
Back-End Development	PHP
Local Development Environment	XAMPP
Database Management System (DBMS)	MySQL/MariaDB
Version Control & Collaboration	Git, GitHub
Testing & Debugging	Postman, Browser Developer Tools
Diagramming & System Design	Draw.io, Canva
PDF Generation	jsPDF

Table 1. Software Requirements for STIzen

Table 1 above summarizes the software requirements for STIzen, highlighting the tools used in each category and their roles in the system's development and operation. Visual Studio Code serves as the primary IDE, supporting coding, debugging, and project management through features like syntax highlighting and error detection, forming the foundation for efficient development.

Front-end and back-end technologies, including HTML5, CSS3, JavaScript, Bootstrap, and PHP, manage the design, responsiveness, and logic of the web-based system. These enable interactive interfaces and core functions such as scheduling, role-based access control, and notification handling, ensuring a smooth user experience across devices and user roles. Bootstrap is specifically utilized to implement a fully responsive layout, ensuring that the system's interface automatically adapts to different screen sizes, including desktops, tablets, and mobile devices, providing a consistent user experience across all platforms.

XAMPP serves as the local development environment, providing an integrated package of Apache, MySQL/MariaDB, and PHP that allows developers to run and test the STIzen system locally before deployment. MySQL/MariaDB serves as the relational database management system, storing schedules, faculty records, student data, and room allocations. Its ACID compliance ensures data integrity, while indexing improves performance and reliability for large datasets. XAMPP provides the local MySQL/MariaDB environment during development and testing, ensuring seamless compatibility between the development and deployment environments. This setup ensures that the system can be efficiently developed, tested, and maintained in a controlled environment before deployment.

Version control and collaboration are managed through Git and GitHub, while Postman and browser developer tools are used for testing and debugging system functionality. Draw.io and Canva support system design visualization, and PDF generation is handled by jsPDF for client-side schedule downloads.

Hardware Requirements

The development, deployment, and operation of STIzen require specific hardware that supports system development, server hosting, and network connectivity. These components ensure the system is reliable, secure, and accessible to all users, while providing a stable environment for development, testing, and deployment.

Hardware	Specification (Minimum / Recommended)
Developer Workstations	Processor: Intel Core i3 or AMD Ryzen 3 RAM: 8 GB / 16 GB Storage: 256 GB SSD / 512 GB SSD
Server Hardware / Hosting Environment	Processor: Dual-core CPU / Quad-core CPU RAM: 4 GB / 8 GB Storage: 128 GB SSD / 256 GB SSD OS: Windows or Linux
Network & Internet Connectivity	Minimum: 10 Mbps wired or wireless Recommended: 50 Mbps wired or wireless
Router / Switch	Wired: Gigabit Ethernet router/switch Wireless: Dual-band 2.4 GHz / 5 GHz router

Table 2. Hardware Requirements for STIzen

Table 2 above summarizes the essential hardware requirements for STIzen, showing the minimum and recommended specifications for development, deployment, and operation. For the development workstation, a processor such as Intel Core i3 or AMD Ryzen 3, 8–16 GB RAM, and 256–512 GB SSD storage ensures that coding, debugging, testing, and documentation can be performed efficiently. This allows developers to run IDEs, local servers, and supporting software smoothly without lag or interruptions. The increase from 8 GB to 16 GB RAM allows smoother execution of multiple development tools and browser instances at the same time. Similarly, upgrading storage from 256 GB to 512 GB SSD provides more space for project files, dependencies, and backup versions.

For the server or hosting environment, a dual-core to quad-core processor, 4–8 GB RAM, and 128–256 GB SSD storage provides the computational power needed to manage multiple concurrent users, handle database transactions, process scheduling requests, and maintain system reliability. The operating system can be either Windows or Linux, providing a stable and secure environment for STIzen deployment. This setup ensures both scalability and high availability for the system. The transition from a dual-core to a quad-core processor enables better handling of simultaneous user requests and background processes. Increasing RAM from 4 GB to 8 GB improves the system's ability to manage larger datasets and more active sessions without slowing down.

For network and internet connectivity, a minimum of 10 Mbps supports basic system access, while a recommended 50 Mbps or higher ensures real-time updates and smooth data synchronization. A Gigabit Ethernet router or switch provides stable wired connections, while a dual-band wireless router supports multiple device connectivity. These hardware components collectively ensure that STIzen operates efficiently, reliably, and with minimal latency for all users. The difference between 10 Mbps and 50 Mbps directly affects how quickly data such as schedules and reports are loaded and updated. The use of dual-band (2.4 GHz and 5 GHz) routers allows better distribution of network traffic across devices, reducing congestion and improving connection stability.

System Requirement Specification

The System Requirements Specification (SRS) outlines the functional and non-functional requirements of the STIzen system. It defines the system's core features, user interactions, and expected behavior, as well as its performance, security, and usability standards, ensuring the system meets stakeholder needs effectively and reliably.

Functional Requirements

Login and Authentication. The system shall provide a Login and Authentication Module that serves as the entry point of STIzen, ensuring that only authorized users can access the platform. The system shall require users to enter their institutional STI email and password, recognizing two domain formats: @naga.sti.edu for administrative roles and @naga.sti.edu.ph for faculty and students. The system shall validate user credentials and, upon successful authentication, direct users to their designated dashboard with access restricted based on their assigned role.

Executive Dashboard. The system shall provide an Executive Dashboard as the main landing page, displaying a role-based overview of data and activities where content dynamically adjusts based on the user's role. For administrative roles (School Administrator, Academic Head, Registrar, and Department Head), the system shall display the Event Board, current Academic Year and Semester, and key statistics such as faculty, schedules, and rooms. For Faculty users, it shall include assigned units, class schedules, examination duties, and teaching load, while for Student users, it shall include the Event Board, Academic Year and Semester, along with class and examination schedules.

Class Schedule. The system shall provide a Class Schedule Module that allows authorized users to create, edit, and delete schedule entries for Senior High School and College programs by assigning faculty, subject, room, and time slot per section, with filtering options by program type, strand or course, and year level. The system shall support section-level locking to prevent further edits and publishing controls to make finalized schedules visible to faculty and students. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Exam Schedule. The system shall provide an Exam Schedule Module that allows authorized users to create, edit, and delete examination schedules for Senior High School and College programs, covering all examination types (preliminary, midterm, pre-final, final, and special examinations). The system shall enable users to set exam dates, assign rooms, and designate faculty as proctors per section, with support for independently scheduling special examinations and applying conflict prevention and controlled visibility of exam information. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Rooms Management. The system shall provide a Rooms Module that allows authorized users to view, add, and update room records, including room name, type, capacity, and availability status. The system shall allow users to toggle room status between Available and Under Maintenance to ensure proper allocation during schedule creation and support conflict-free room assignment across SHS and College operations with real-time updates. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Faculty Management. The system shall provide a Faculty Module that allows authorized users to add, edit, and manage faculty records, including department assignments and activation status. The system shall ensure that only active and properly assigned faculty members are available for schedule creation. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Subjects Management. The system shall provide a Subjects Module that allows authorized users to add, update, and manage subjects categorized by strand or course and year level. The system shall ensure that only valid and curriculum-aligned subjects are used during schedule creation. The system shall validate subject details to prevent duplicate or incomplete entries. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Course & Section Management. The system shall provide a Course & Section Module that allows authorized users to add, edit, and manage SHS strands, College courses, and their respective sections, including options to flag merged sections with corresponding notes. The system shall display section overviews with details such as program type, merge status, subject count, and estimated student count, and support search and filtering by department. The system shall also manage Academic Year and Term settings used as scheduling references across the system. Access to these functions shall be restricted based on user roles and permissions defined in the system.

Users Management. The system shall provide a User Module that allows the Academic Head to manage all user accounts in STIzen, organized into Faculty and Student tabs. The system shall enable the Administrator to add, edit, activate, and deactivate Faculty accounts (including faculty members, department heads, registrars, and other staff roles) and Student accounts, with search and filtering options by role, course, and section. The system shall enforce role-based access control by assigning permissions that determine each user's accessible modules and actions.

Schedule Versions. The system shall provide a Schedule Versions Module that enables the Academic Head, Registrar, and Department Head to view the complete history of all schedule changes. The system shall record each version with a timestamp, change description, and the user responsible, and shall allow authorized users to compare different versions and restore previous schedules when necessary by accessing stored schedule snapshots, with a maximum of 50 versions retained per schedule to manage system storage and maintain performance.

Audit Logs. The system shall provide an Audit Logs Module that enables the Academic Head and School Administrator to monitor all system activities. The system shall record each action with details such as the user, action type, and timestamp, and shall provide filtering options by user, action type, and date range to support efficient tracking, review, and investigation of system activities.

Event Board. The system shall provide an Event Board Module that allows the Academic Head to create, edit, and delete school-wide announcements by posting event details such as title, description, and optional images. The system shall display these announcements to all user roles as a centralized communication channel, ensuring that important school updates and announcements are consistently disseminated across the institution.

Notifications. The system shall provide a Notifications Module that delivers real-time, role-based alerts to users regarding important scheduling activities and system events. The system shall automatically generate notifications for schedule creation, updates, publishing, faculty assignments, exam proctoring, and announcements. Notifications shall be displayed within the system interface and dashboard to ensure visibility. Each notification shall be tailored to the user's role to ensure that only relevant information is received. This module enhances communication and supports timely coordination by keeping all users informed of schedule changes and system updates.

Non-Functional Requirements

Security. The system must enforce role-based access control to ensure that each user can only access modules and actions permitted for their assigned role. The system must use secure authentication through institutional STI email credentials, and sensitive data such as passwords must be securely stored and protected against unauthorized access, disclosure, or tampering.

Usability. The system must provide a clean and intuitive interface that is easy to use for all user roles without requiring extensive technical knowledge. Navigation must be role-based, displaying only relevant options per user, and core functions such as schedule creation and record management must be simple and efficient to perform.

Reliability. The system must operate consistently across all modules without unexpected failures or data loss. Critical processes such as schedule creation, publishing, locking, and version restoration must function accurately and consistently to ensure data integrity throughout the academic period. The system must also maintain stable performance during continuous use and prevent disruptions that may affect ongoing academic operations.

Performance. The system must load pages and process user actions efficiently, even when handling large volumes of scheduling data. Operations such as filtering schedules, generating dashboard statistics, and retrieving audit logs must respond promptly to avoid delays in daily administrative workflows.

Maintainability. The system must be structured to allow easy updates, bug fixes, and feature additions without disrupting existing functionality. Its modular architecture supports targeted maintenance, and the availability of audit logs and schedule version history aids in diagnosing and resolving issues efficiently.

Scalability. The system must be capable of accommodating a growing number of users, records, and scheduling entries without significant impact on performance. The system must also support the addition of new strands, courses, sections, and user roles as the institution's academic offerings expand over time. It must be designed to handle increased system load efficiently while maintaining consistent responsiveness and functionality.

Accessibility. The system must be operable through standard web browsers on any device, including desktops, tablets, and mobile phones, without requiring additional software or plugins. The interface must be fully responsive and adaptive across different screen sizes, ensuring that all layout elements, navigation menus, tables, and forms automatically adjust to fit the user's screen. The interface must be clearly legible and well-organized, ensuring that all user roles can access the information and functions they need without difficulty.

System Evaluation (Tools and Methods)

The evaluation of the system was conducted to determine its effectiveness, usability, and alignment with the operational needs of STI College Naga. The evaluation focused on gathering insights from end users and key stakeholders, assessing system functionality, and identifying areas for improvement.

To achieve this, the evaluation involved two groups of participants to gather comprehensive feedback on the system. The first group consisted of key personnel, including the Administrator, Registrar, Academic Head, Department Heads, and selected Faculty members. These participants were interviewed directly in person following a formal request and approval process. The purpose of these interviews was to capture detailed insights into current scheduling practices, identify pain points, and understand the functionalities they desire in the new system.

The second group comprised students, with a total of 152 participants representing various programs and sections, which reflects a cross-section of the total student population of 599. The student evaluation focused on assessing the system's accessibility and usability across different devices, identifying preferred features, and understanding notification preferences related to class and exam scheduling. Overall, the combined feedback from both groups provided a holistic view of the system’s strengths and areas for improvement. By listening to both staff and students, the researchers ensured that the system worked effectively for all users across different devices and screen sizes, helping create a solution that meets the needs of the entire school community.

Data Collection Methods

The data collection process for evaluating the system utilized multiple methods and supporting tools to ensure comprehensive feedback. To begin with, structured interviews were conducted with key personnel to gain a deeper understanding of challenges in the existing manual or partially automated scheduling system, providing qualitative insights on requirements, faculty load management, conflict resolution, and reporting needs. In addition, surveys were administered to students to assess system usability, feature preferences, and accessibility, collecting both quantitative and qualitative feedback through Google Forms.

Furthermore, observation and hands-on testing were conducted, where faculty and administrators interacted directly with the STIzen web system, while students evaluated the system's web portal on various devices. As a result, researchers were able to assess navigation, functionality, and the accuracy of schedule generation in real time, identify usability challenges, and verify that the system effectively met the institution's operational requirements.

Tools Used in Evaluation

To evaluate the system, tools and methods were used to collect and analyze data from stakeholders. The table below summarizes these tools and their purposes in the assessment. These were selected to ensure accurate and clear data interpretation. Using multiple methods provided a comprehensive understanding of the system’s performance in real-world scenarios.

Tool / Method	Purpose / Functionality
Google Forms	Used to administer surveys and automatically collect, organize, and summarize student responses for analysis
Face-to-Face Interviews	Conducted to gather in-depth qualitative insights from administrators and key personnel regarding system needs, challenges, and feedback
Observation / Prototype Testing	Used to evaluate system usability, navigation, and accuracy through hands-on interaction with wireframes and system prototypes

Table 3. Tools Used in System Evaluation

Table 3 above highlights the key tools and methods employed to evaluate STIzen. It presents an overview of how each tool contributed to assessing system usability, functionality, and user interaction. These tools collectively ensured a comprehensive and reliable evaluation of the system from both administrative and student perspectives. This approach also strengthened the validity of the results by allowing cross-verification of data from multiple sources. Google Forms enabled efficient collection and automatic organization of survey responses, making quantitative data easier to analyze. Face-to-face interviews and observation/prototype testing provided detailed qualitative feedback by allowing direct interaction with administrators and hands-on evaluation of system features. It also ensured that both technical performance and user experience were properly assessed before final implementation.

Data Handling & Confidentiality

All data collected during the evaluation were handled with strict confidentiality and used solely for the purpose of assessing and improving the STIzen system for this Capstone project. Participation was entirely voluntary, allowing students and personnel to opt out of any questions they preferred not to answer. The information gathered from interviews, questionnaires, and prototype observations was carefully analyzed to inform system refinements, ensuring that STIzen effectively addresses the real-world needs of STI College Naga's stakeholders while maintaining privacy and ethical standards. Furthermore, all collected data were securely stored and accessible only to the research proponents to prevent unauthorized access. No personally identifiable information was disclosed in any part of the study, ensuring that all results were presented in an aggregated and anonymous manner.

Evaluation Method and Research Instrument

To assess the effectiveness and technical quality of the system, the researchers adopted the ISO/IEC 25010 Software Quality Model, a widely recognized international standard for software evaluation. This model assesses software based on quality characteristics such as functionality, reliability, usability, performance efficiency, maintainability, security, compatibility, and portability, ensuring that the system meets established software engineering standards.

To operationalize this evaluation, a structured questionnaire was used as the primary research instrument, with each item aligned to the relevant ISO/IEC 25010 quality characteristics. Respondents, including key stakeholders at STI College Naga, evaluated each item using a 5-point Likert Scale ranging from 1 (Strongly Disagree/Very Poor) to 5 (Strongly Agree/Excellent), allowing for quantitative assessment of system quality across all dimensions.

Scale	Range	Verbal Interpretation
5	4.21 – 5.00	Strongly Agree (Excellent)
4	3.41 – 4.20	Agree (Very Good)
3	2.61 – 3.40	Neutral (Good)
2	1.81 – 2.60	Disagree (Poor)
1	1.00 – 1.80	Strongly Disagree (Very Poor)

Table 4. The 5-Point Likert Scale and Verbal Interpretation

Table 4 above shows the 5-point Likert scale used in the evaluation, along with its corresponding verbal interpretations. This scale allows respondents to rate each survey item based on the ISO/IEC 25010 Software Quality Model, enabling the assessment of system quality across key characteristics such as usability, reliability, and performance.

Each verbal interpretation reflects the level of user satisfaction and system quality. Excellent indicates that the system fully meets or exceeds user expectations with optimal performance and no significant issues. Very Good means the system performs well with only minor issues that do not affect usability. Good suggests acceptable functionality but with noticeable areas for improvement. Poor indicates several weaknesses that impact user experience and require enhancement, while Very Poor means the system does not meet user requirements and needs major improvement or redesign.

Diagrams

To provide a clear understanding of the structure, flow, and interactions of the STIzen Academic Scheduling Management System, the following diagrams were created. These diagrams visually represent the relationships between data, the flow of information within the system, and the interactions between users and system functionalities. All diagrams were designed using draw.io, ensuring clarity, consistency, and ease of interpretation for both stakeholders and developers.

Entity-Relationship Diagram (ERD)

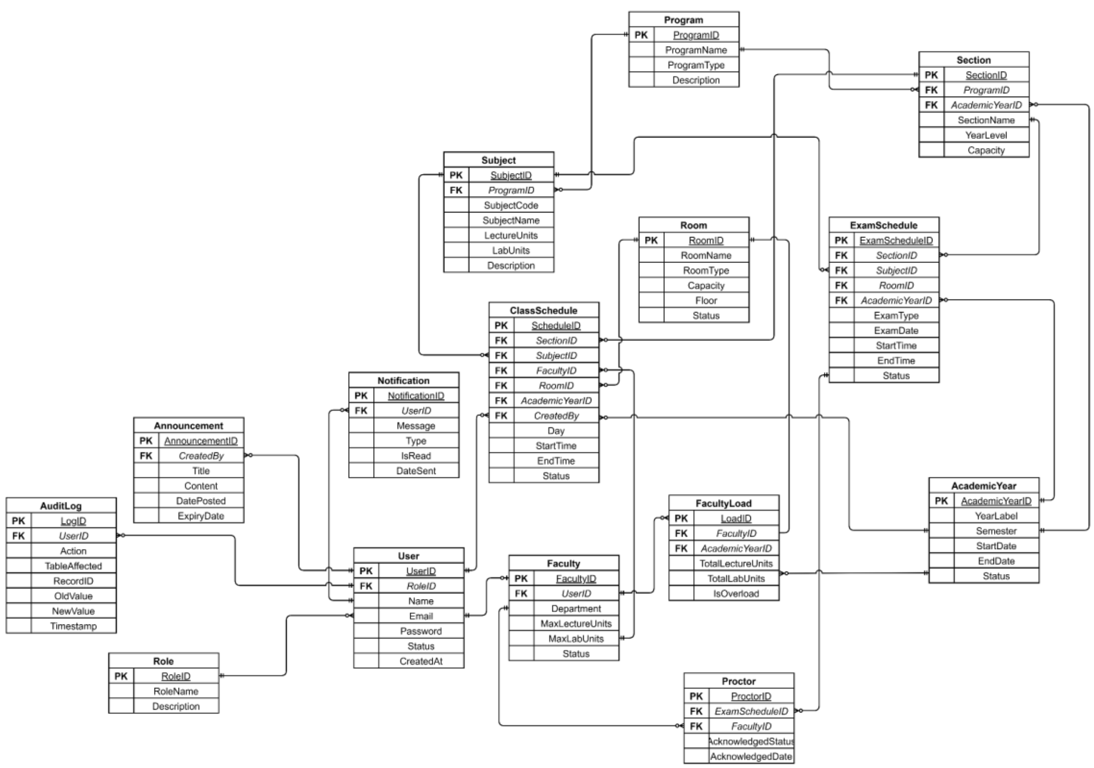

Figure 3. Entity-Relationship Diagram (Crows' Foot Notation) of the STIzen System

Figure 3 above presents the Entity-Relationship Diagram (ERD) of STIzen system, illustrating the database structure and relationships among its fifteen (15) entities. Each entity is represented as a table with attributes, primary keys (PK) underlined, and foreign keys (FK) italicized to indicate key constraints. Relationship cardinalities are shown using crow's foot notation, where a single vertical bar represents "one" and a crow's foot represents "many."

In addition, the User entity serves as the central reference point, connecting to Role, Faculty, Notification, Announcement, and Audit Log. These links indicate that a user has one role, is associated with a faculty record, receives many notifications, creates announcements, and generates log entries. This central connection ensures consistent tracking of user activities across the system.

Moreover, the ERD demonstrates how scheduling and academic entities interact. Faculty connects to ClassSchedule, FacultyLoad, and Proctor, while Program links to Subject and Section. Both ClassSchedule and ExamSchedule reference multiple foreign keys, with Proctor bridging ExamSchedule and Faculty, reflecting accurate scheduling assignments and proctoring responsibilities.

Data Flow Diagram (DFD)

The DFD depicts the movement of data through STIzen, highlighting inputs, processes, and outputs. It shows how users interact with the system, how scheduling requests are processed, and how reports, notifications, and updates are generated. This diagram helps in understanding the system's functional processes and the flow of information between modules.

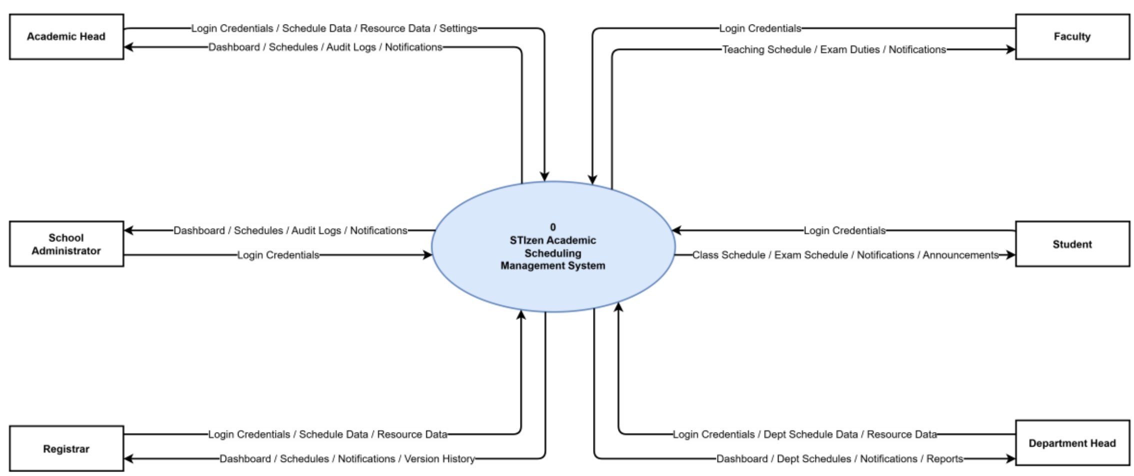

Figure 4. STIzen DFD Context Diagram (Level 0) — Yourdon and De Marco Notation

Figure 4 above presents the highest-level view of the STIzen system, represented as a single process bubble labeled "0 — STIzen Academic Scheduling Management System." It defines the system boundary and illustrates interactions with six external entities through labeled data flows, without revealing internal logic. The School Administrator, Academic Head, Registrar, Department Head, Faculty, and Students provide inputs such as approvals, schedules, and updates, and receive outputs including reports, notifications, and PDFs.

Each external entity interacts based on its specific role, ensuring efficient flow of tasks and information across the system. As a result, the diagram establishes the complete input-output interface of the system, serving as a clear foundation for understanding external relationships before examining internal processes. This overview helps stakeholders quickly identify how data enters and exits the system. It also supports better analysis and design of subsequent, more detailed system diagrams.

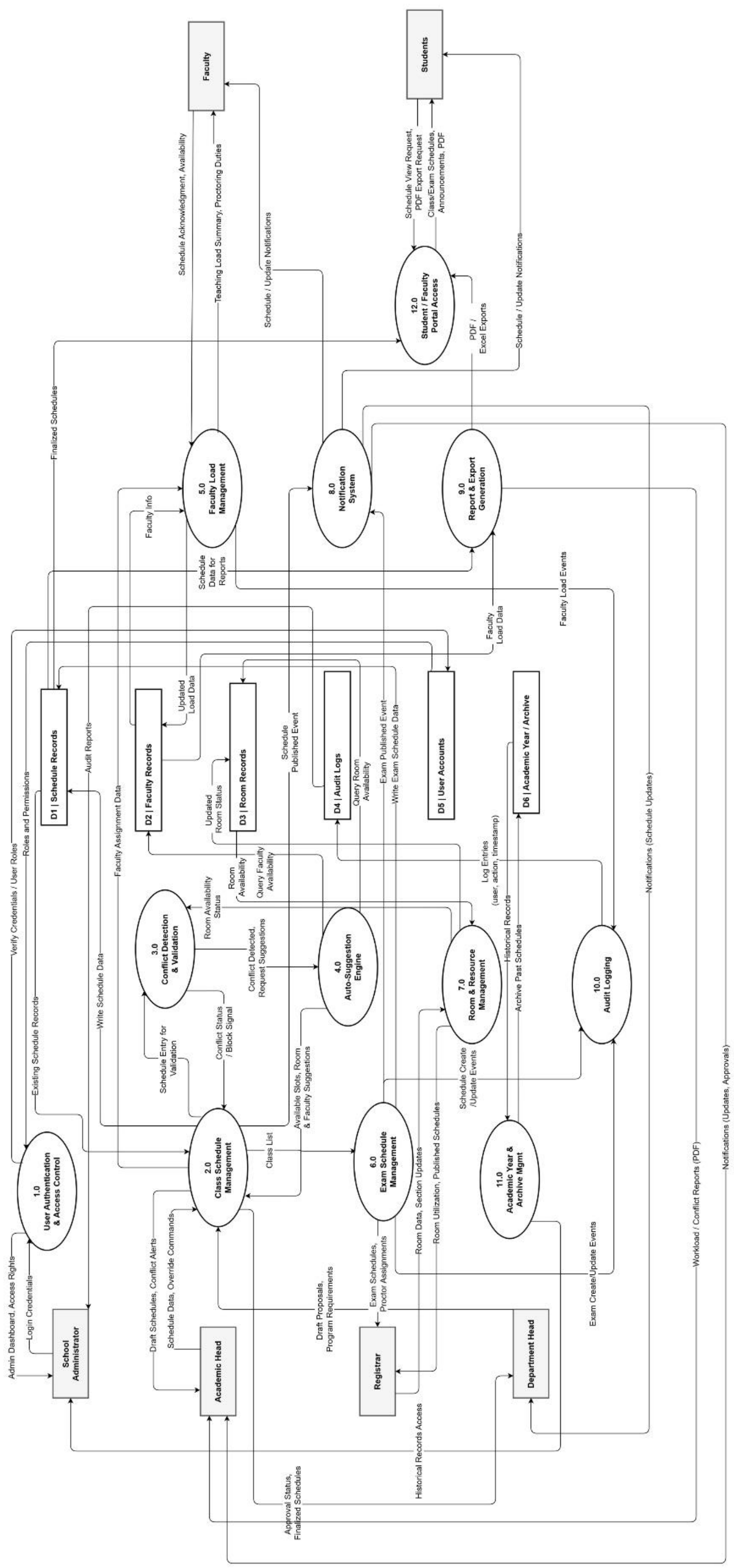

Figure 5. STIzen DFD Level 1 — Yourdon and De Marco Notation

Figure 5 above expands the single process from the Context Diagram into twelve major sub-processes, representing the core functional modules of the STIzen system. This decomposition introduces six data stores, labeled D1 through D6, while maintaining connections to all six external entities. By providing a detailed functional map, the diagram preserves an appropriate level of abstraction, allowing stakeholders to understand the system's operations without exposing excessive internal complexity. For example, Process 1.0 User Authentication and Access Control validates credentials and enforces role-based permissions, while Process 2.0 Class Schedule Management handles schedule creation, editing, approval, locking, and publication.

Furthermore, the diagram illustrates processes that enhance efficiency and prevent conflicts within the system. Process 3.0 Conflict Detection and Validation automatically identifies faculty, room, and section overlaps using preventive validation, whereas Process 4.0 Auto-Suggestion Engine provides intelligent recommendations for available time slots, rooms, and faculty assignments. Process 5.0 Faculty Load Management monitors teaching loads, issues overload warnings, and enables digital acknowledgment, while Process 6.0 Exam Schedule Management oversees exam creation, room allocation, proctor assignments, and conflict validation. Process 7.0 Room and Resource Management tracks room availability and utilization, and Process 8.0 Notification System delivers timely, role-based alerts triggered by key system events such as schedule publication or overload conditions.

Finally, the DFD highlights processes for reporting, auditing, archival, and portal access. Process 9.0 Report and Export Generation produces reports on faculty workloads, conflicts, room usage, and exam schedules in PDF or Excel formats. Process 10.0 Audit Logging captures all user actions with timestamps to maintain accountability. Process 11.0

Academic Year and Archive Management manages academic term structures and archives past schedule data as read-only records, while Process 12.0 Student and Faculty Portal Access enables authenticated users to view finalized schedules, announcements, and exportable PDFs. The six data stores—D1 Schedule Records, D2 Faculty Records, D3 Room Records, D4 Audit Logs, D5 User Accounts, and D6 Academic Year and Archive—provide a persistent data layer to ensure system reliability, historical tracking, and secure management of all academic scheduling information.

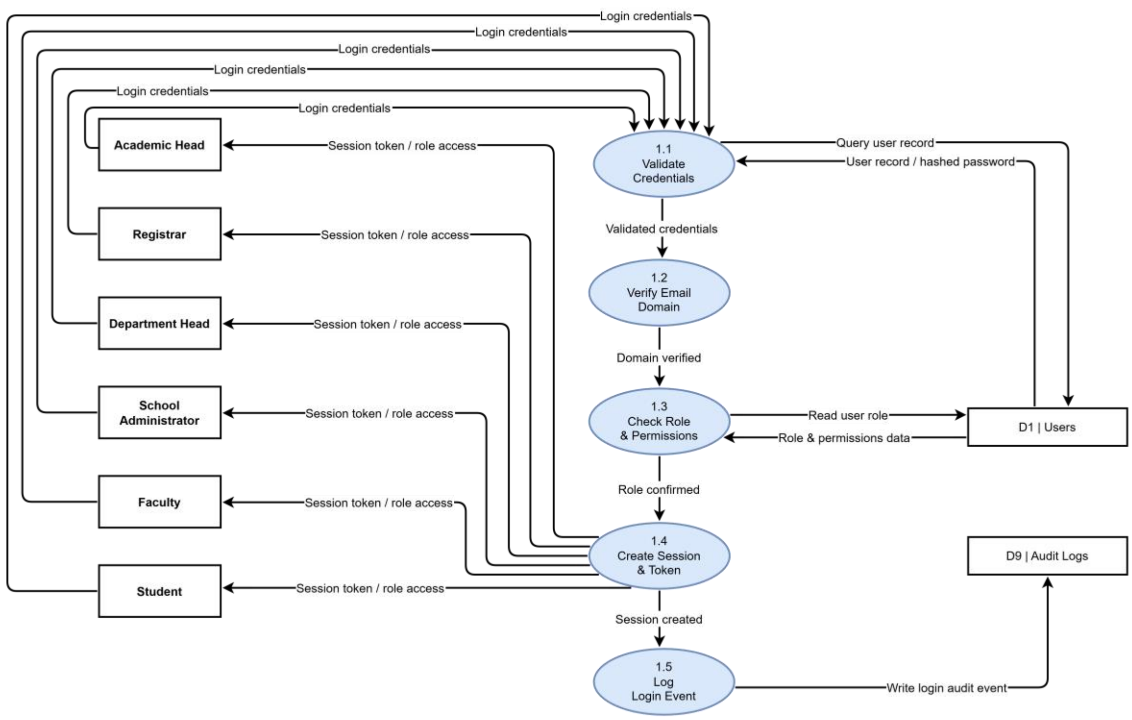

Figure 6. STIzen DFD Level 2 — Process 1.0 Login and Authentication

Figure 6 above illustrates the Login and Authentication process of the STIzen system, serving as the entry point for six user types: Academic Head, Registrar, Department Head, School Administrator, Faculty, and Student. The process consists of five sequential sub-processes forming an authentication pipeline. All users submit their credentials to sub-process 1.1 (Validate Credentials), which checks the Users data store (D1) for matching records and hashed passwords. The validated data is then passed to sub-process 1.2 (Verify Email Domain) to ensure the email belongs to an authorized institutional domain.

Once verified, the credentials proceed to sub-process 1.3 (Check Role and Permissions), which retrieves role and access rights from D1. Sub-process 1.4 (Create Session and Token) then generates a unique session token and grants dashboard access. Finally, sub-process 1.5 (Log Login Event) records the login activity in the Audit Logs data store (D9).

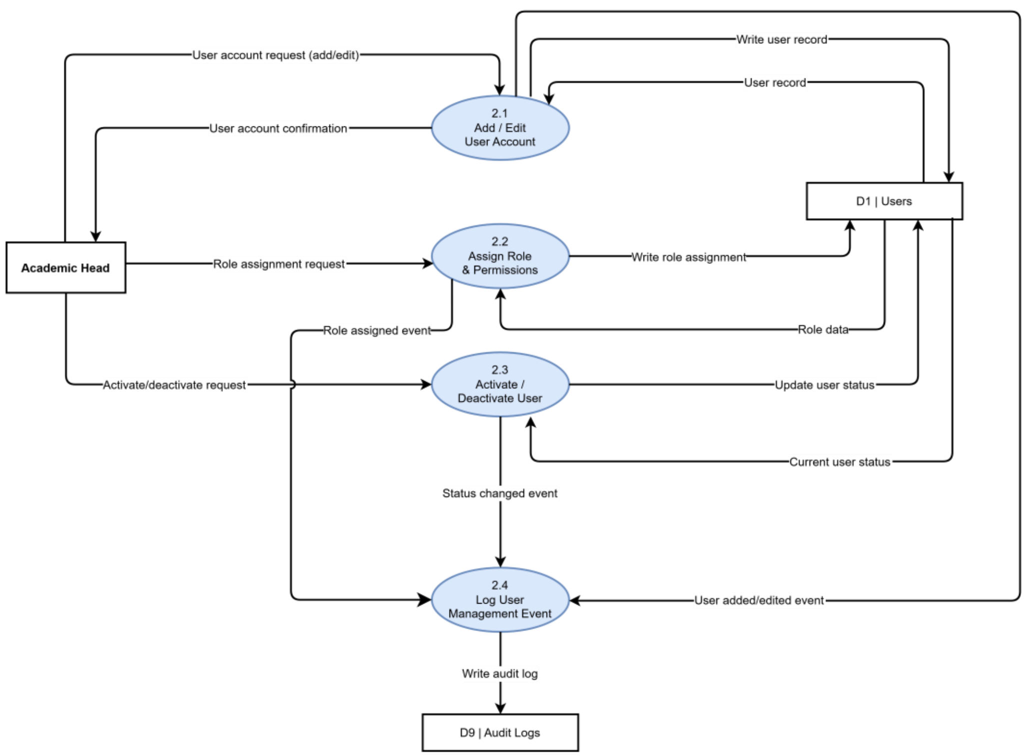

Figure 7. STIzen DFD Level 2 — Process 2.0 Manage Users

Figure 7 above illustrates the Manage Users process of the STIzen system, handled exclusively by the Academic Head as the authorized entity for user account management.

The process is divided into four sub-processes. Sub-process 2.1 (Add/Edit User Account) processes account creation or updates, stores the data in the Users data store (D1), and returns a confirmation while forwarding the event for logging. Sub-process 2.2 (Assign Role and Permissions) manages role assignments by updating and verifying role data in D1, then sends the event to the logging process.

Sub-process 2.3 (Activate/Deactivate User) handles account status changes by updating user status in D1 and forwarding the status change event. Finally, sub-process 2.4 (Log User Management Event) consolidates all events and records them in the Audit Logs data store (D9) to ensure traceability of all user management actions.

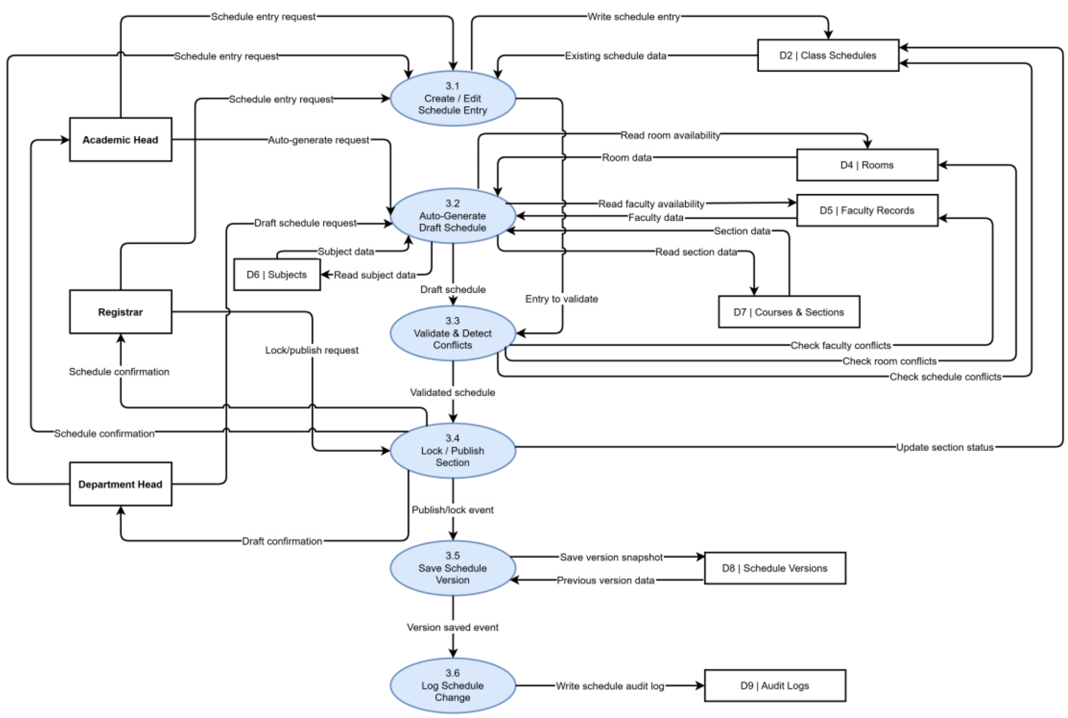

Figure 8. STIzen DFD Level 2 — Process 3.0 Manage Class Schedules

Figure 8 above illustrates the Manage Class Schedules process of the STIzen system, involving the Academic Head, Registrar, and Department Head, each with distinct roles in scheduling. Sub-processes 3.1 (Create/Edit Schedule Entry) and 3.2 (Auto-Generate Draft Schedule) handle manual and automated schedule creation using data from Class Schedules (D2), Rooms (D4), Faculty Records (D5), Subjects (D6), and Courses and Sections (D7), with all outputs sent to sub-process 3.3 (Validate and Detect Conflicts) for verification.

Once validated, sub-process 3.4 (Lock/Publish Section) finalizes and confirms schedules, sub-process 3.5 (Save Schedule Version) stores version snapshots in D8, and sub-process 3.6 (Log Schedule Change) records all changes in the Audit Logs data store (D9) for traceability.

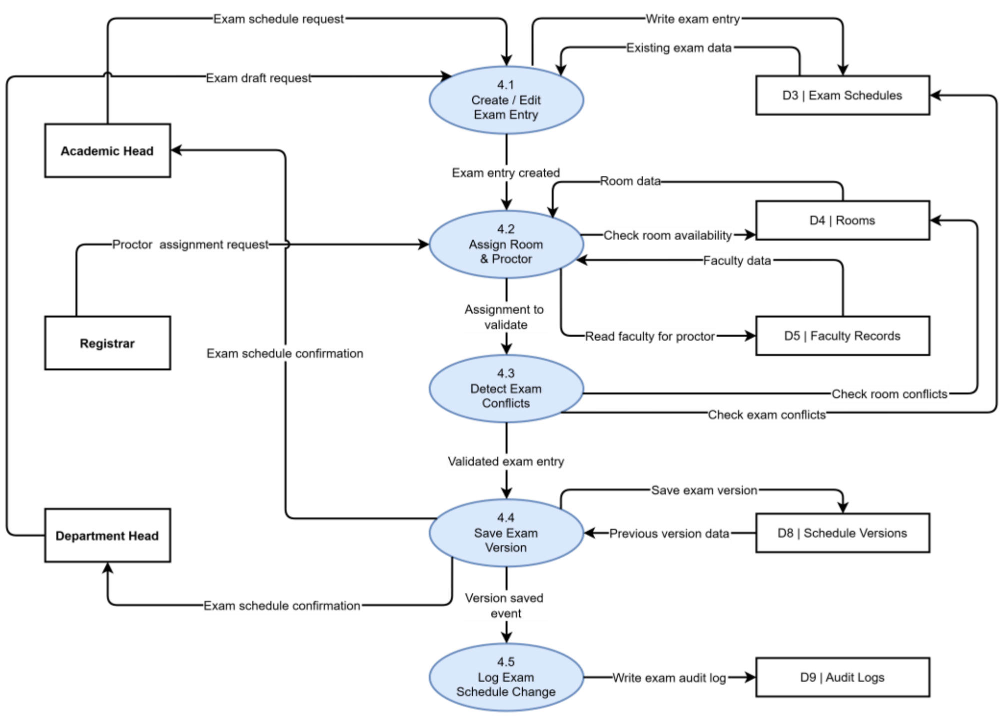

Figure 9. STIzen DFD Level 2 — Process 4.0 Manage Exam Schedules

Figure 9 above illustrates the Manage Exam Schedules process of the STIzen system, involving the Academic Head, Registrar, and Department Head. The process includes sub-processes 4.1 (Create/Edit Exam Entry) and 4.2 (Assign Room and Proctor), which handle exam creation, room allocation, and proctor assignment using data from Exam Schedules (D3), Rooms (D4), and Faculty Records (D5). Sub-process 4.3 (Detect Exam Conflicts) validates entries by checking scheduling and room conflicts before passing them forward.

Once validated, sub-process 4.4 (Save Exam Version) stores a version snapshot in D8 and returns confirmation to users, while sub-process 4.5 (Log Exam Schedule Change) records all updates in the Audit Logs data store (D9) to ensure traceability.

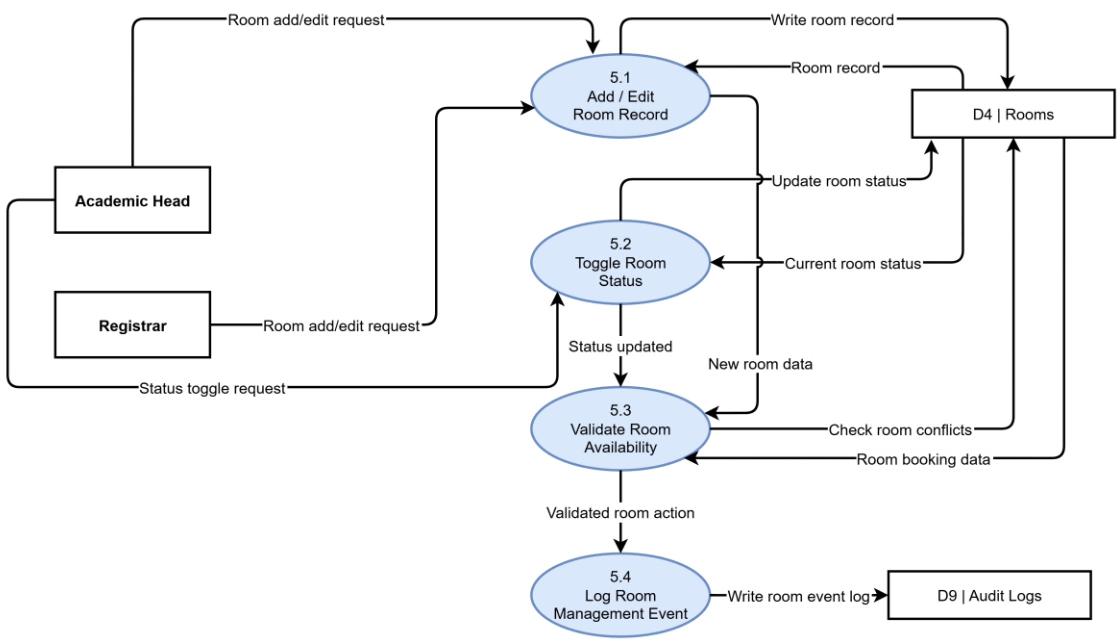

Figure 10. STIzen DFD Level 2 — Process 5.0 Manage Rooms

Figure 10 above illustrates the Manage Rooms process of the STIzen system, involving the Academic Head and Registrar as authorized entities for managing room records. The process includes four sub-processes. Sub-process 5.1 (Add/Edit Room

Record) handles the creation and updating of room data by writing to and retrieving records from the Rooms data store (D4), while sub-process 5.2 (Toggle Room Status) manages room activation or deactivation by updating the room status in D4.

Both updates are forwarded to sub-process 5.3 (Validate Room Availability), which checks existing room bookings in D4 to prevent conflicts or double scheduling. Once validated, sub-process 5.4 (Log Room Management Event) records all room-related actions in the Audit Logs data store (D9), ensuring full traceability of changes within the system.

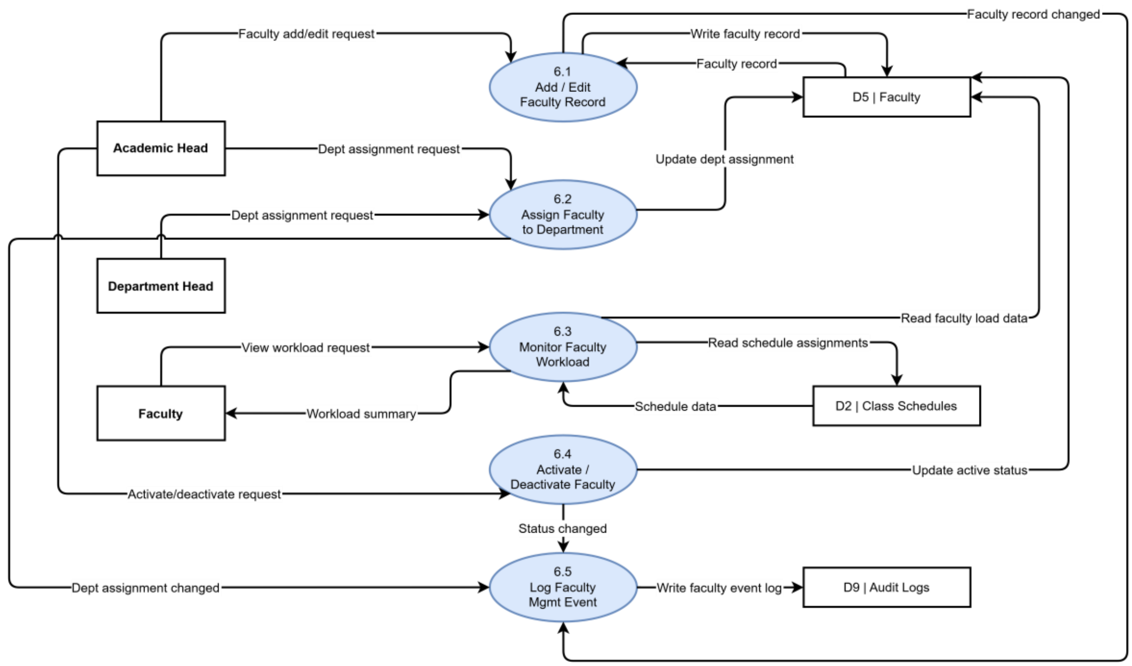

Figure 11. STIzen DFD Level 2 — Process 6.0 Manage Faculty

Figure 11 above illustrates the Manage Faculty process of the STIzen system, involving the Academic Head, Department Head, and Faculty members. The process consists of five sub-processes. Sub-process 6.1 (Add/Edit Faculty Record) manages the creation and updating of faculty data in the Faculty data store (D5), while sub-process 6.2

(Assign Faculty to Department) handles department assignments by updating records in D5. Sub-process 6.3 (Monitor Faculty Workload) allows faculty members to view their workload by retrieving data from both D5 and Class Schedules (D2) to generate a workload summary.

Sub-process 6.4 (Activate/Deactivate Faculty) updates the active status of faculty records in D5 and forwards the event for logging. Finally, sub-process 6.5 (Log Faculty Management Event) consolidates all changes and records them in the Audit Logs data store (D9), ensuring traceability of all faculty-related actions.

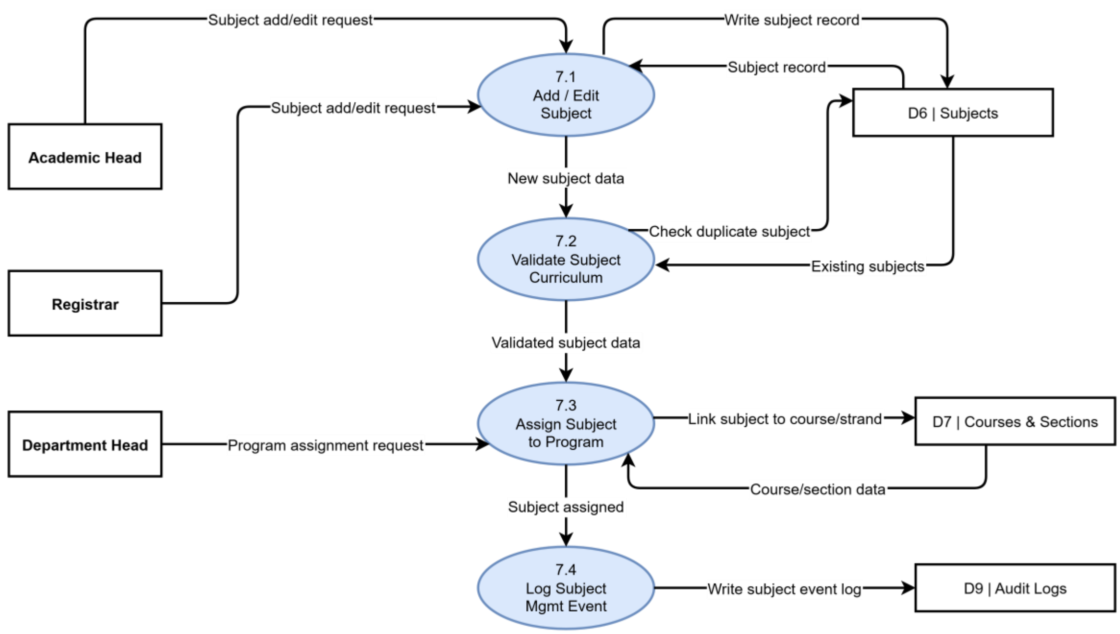

Figure 12. STIzen DFD Level 2 — Process 7.0 Manage Subjects

Figure 12 above illustrates the Manage Subjects process of the STIzen system, involving the Academic Head, Registrar, and Department Head. The process consists of four sub-processes. Sub-process 7.1 (Add/Edit Subject) handles the creation and updating of subject records in the Subjects data store (D6), which are then forwarded to sub-process 7.2 (Validate Subject Curriculum) to check for duplicates and ensure data validity.

Once validated, sub-process 7.3 (Assign Subject to Program) links subjects to the appropriate courses or strands by updating the Courses and Sections data store (D7). Finally, sub-process 7.4 (Log Subject Management Event) records all subject-related actions in the Audit Logs data store (D9), ensuring full traceability within the system.

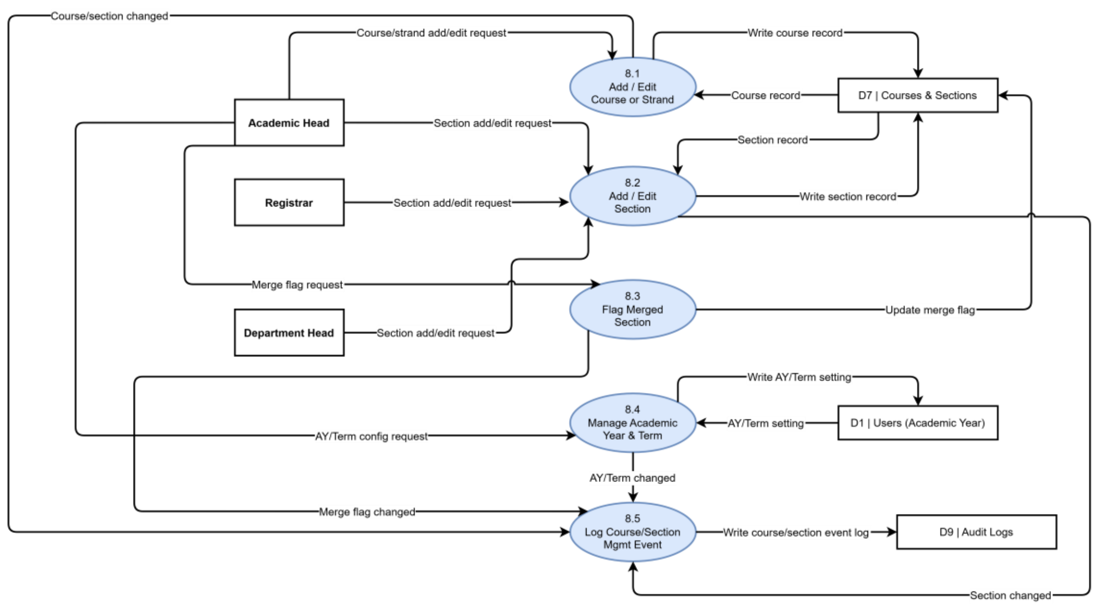

Figure 13. STIzen DFD Level 2 — Process 8.0 Manage Course and Sections

Figure 13 above illustrates the Manage Course and Sections process of the STIzen system, involving the Academic Head, Registrar, and Department Head. The process is composed of five sub-processes. Sub-process 8.1 (Add/Edit Course or Strand) and 8.2 (Add/Edit Section) handle the creation and updating of course and section records in the Courses and Sections data store (D7), while sub-process 8.3 (Flag Merged Section) updates section merge status when needed.

Sub-process 8.4 (Manage Academic Year and Term) configures academic year and term settings by updating the Users data store (D1). Finally, sub-process 8.5 (Log Course/Section Management Event) consolidates all changes and records them in the Audit Logs data store (D9), ensuring full traceability of all course and section management activities.

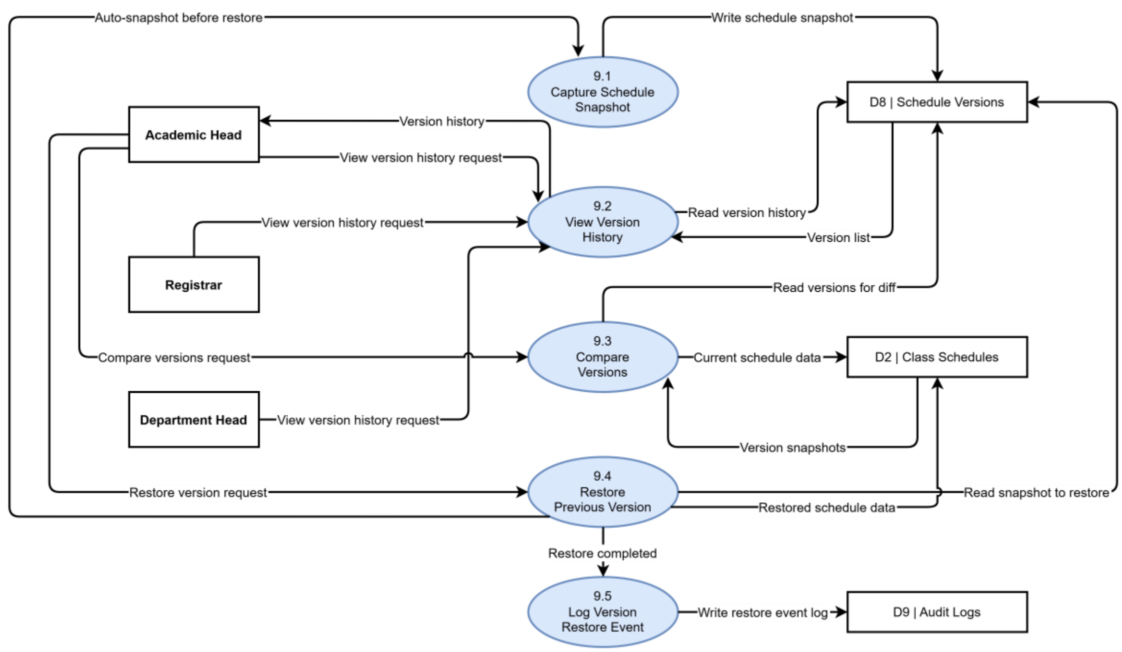

Figure 14. STIzen DFD Level 2 — Process 9.0 Manage Schedule Versions

Figure 14 above illustrates the Manage Schedule Versions process of the STIzen system, accessible to the Academic Head, Registrar, and Department Head. The process includes five sub-processes. Sub-process 9.1 (Capture Schedule Snapshot) automatically saves schedule states to the Schedule Versions data store (D8), while sub-process 9.2 (View Version History) allows users to retrieve and review saved versions from D8. Sub-process 9.3 (Compare Versions) enables comparison of schedule snapshots with current data from Class Schedules (D2) for analysis.

Sub-process 9.4 (Restore Previous Version) restores selected snapshots to D2 while first capturing a backup of the current state. Finally, sub-process 9.5 (Log Version Restore Event) records restoration actions in the Audit Logs data store (D9), ensuring full traceability of version control activities.

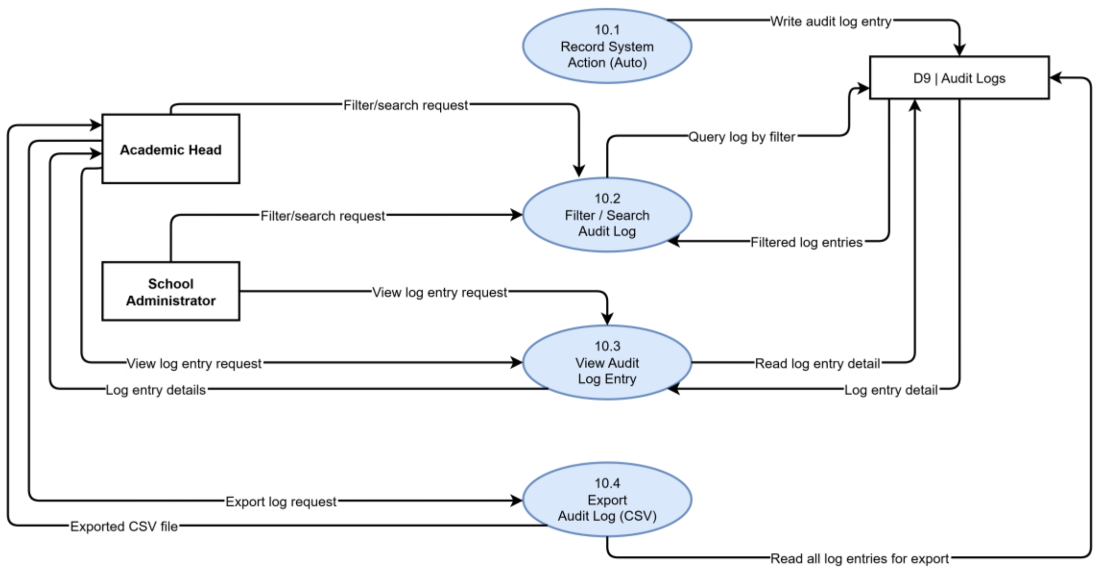

Figure 15. STIzen DFD Level 2 — Process 10.0 Manage Audit Logs

Figure 15 above illustrates the Manage Audit Logs process of the STIzen system, involving the Academic Head and School Administrator as authorized users for accessing audit data. The process is composed of four sub-processes. Sub-process 10.1 (Record System Action – Auto) automatically captures and writes audit log entries to the Audit Logs data store (D9) whenever any system action occurs, such as login events, schedule updates, or user management activities. Sub-process 10.2 (Filter/Search Audit Log) allows users to query and filter log entries from D9 based on specific criteria, while sub-process 10.3 (View Audit Log Entry) retrieves and displays detailed information for a selected log record. Sub-process 10.4 (Export Audit Log – CSV) enables the Academic Head to export all audit logs from D9 into a CSV file for reporting and offline use, ensuring complete transparency and recordkeeping of system activities.

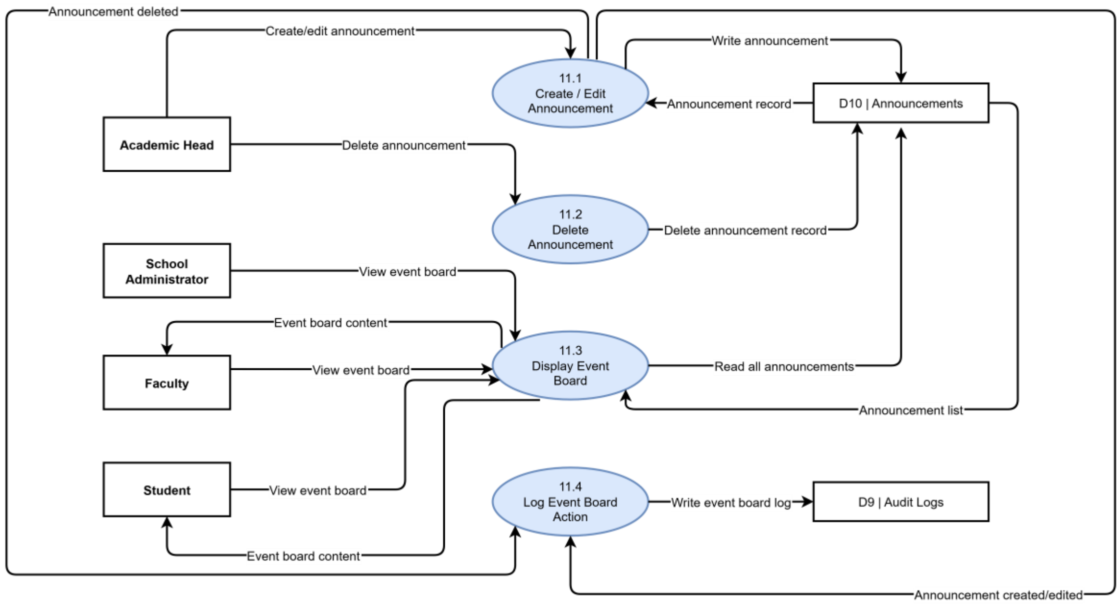

Figure 16. STIzen DFD Level 2 — Process 11.0 Manage Event Board

Figure 16 above illustrates the Manage Event Board process of the STIzen system, involving the Academic Head and School Administrator as announcement managers, and Faculty and Students as primary viewers. The process consists of four sub-processes. Sub-process 11.1 (Create/Edit Announcement) handles the creation and updating of announcements in the Announcements data store (D10), while sub-process 11.2 (Delete Announcement) removes selected announcements from D10 when requested.

Sub-process 11.3 (Display Event Board) retrieves all active announcements from D10 and displays them to Faculty and Students for system-wide viewing. Finally, sub-process 11.4 (Log Event Board Action) records all creation, modification, and deletion activities in the

Audit Logs data store (D9), ensuring that all event board actions are properly documented and traceable.

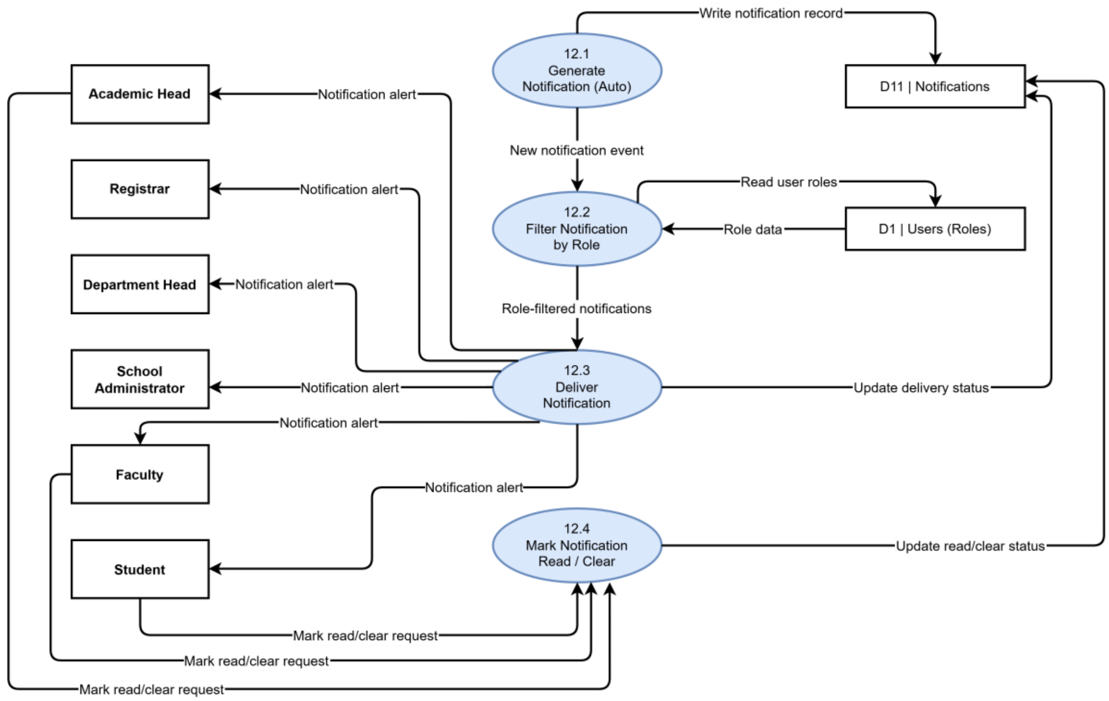

Figure 17. STIzen DFD Level 2 – Process 12.0 Manage Notifications

Figure 17 above illustrates the Manage Notifications process of the STIzen system, involving all six external entities: Academic Head, Registrar, Department Head, School Administrator, Faculty, and Students. The process consists of four sub-processes. Sub-process 12.1 (Generate Notification – Auto) automatically creates notifications triggered by system events such as schedule updates, exam changes, or conflict resolutions, and stores them in the Notifications data store (D11).

Sub-process 12.2 (Filter Notification by Role) processes these notifications by retrieving user role information from the Users data store (D1) to determine the appropriate recipients. Sub-process 12.3 (Deliver Notification) then distributes the filtered notifications to the corresponding users and updates their delivery status in D11. Finally, sub-process 12.4 (Mark Notification Read/Clear) allows users to acknowledge or dismiss notifications, updating their status in D11 to ensure accurate tracking of notification states across the system.

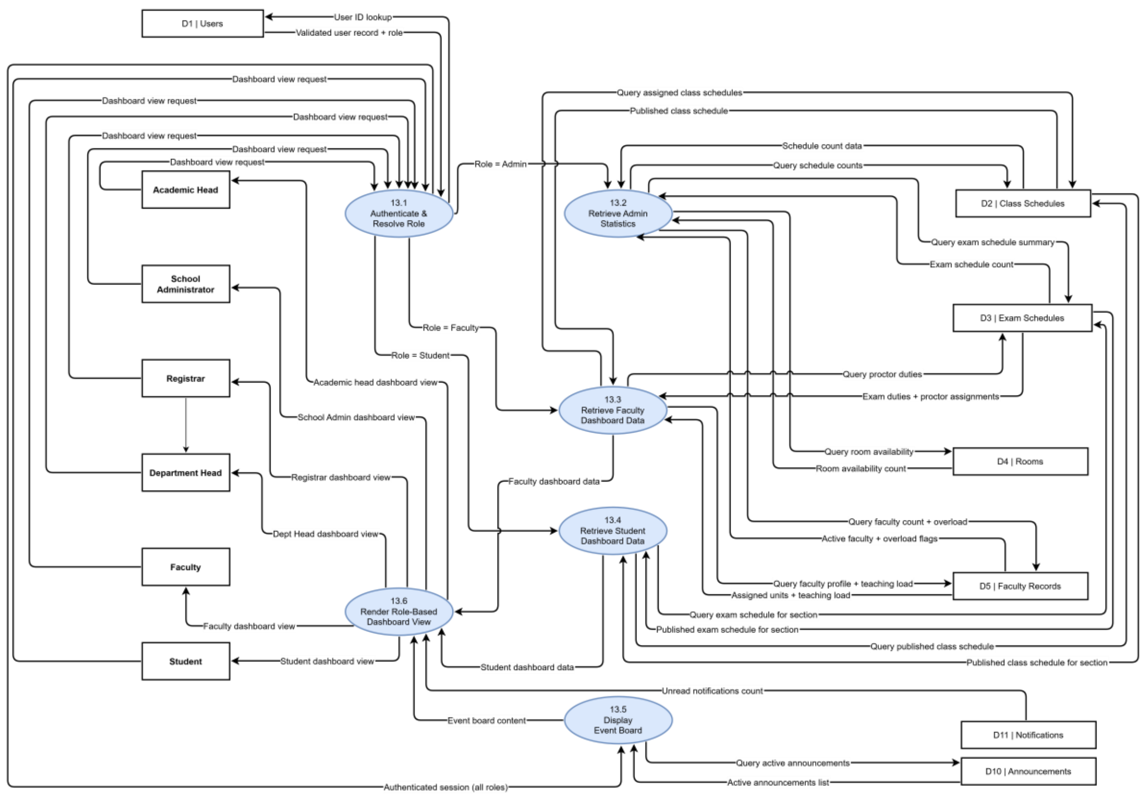

Figure 18. STIzen DFD Level 2 – Process 13.0 Executive Dashboard

Figure 18 above illustrates the Executive Dashboard process of the STIzen system, which serves as the centralized role-based landing interface for Academic Head, School Administrator, Registrar, Department Head, Faculty, and Students. The process is composed of six sub-processes that authenticate users, retrieve role-specific data, and generate a personalized dashboard. Sub-process 13.1 (Authenticate and Resolve Role) validates users through the Users data store (D1) and determines their assigned roles. Based on this role, requests are routed to the appropriate data retrieval processes while all users are also directed to view the event board.

Sub-process 13.2 (Retrieve Admin Statistics) gathers system-wide data for admin users from Class Schedules (D2), Exam Schedules (D3), Rooms (D4), and Faculty Records (D5). Sub-process 13.3 (Retrieve Faculty Dashboard Data) collects faculty profile, workload, class schedules, and exam assignments from D5, D2, and D3. Sub-process 13.4 (Retrieve Student Dashboard Data) retrieves class and exam schedules for students from D2 and D3. All retrieved data from these sub-processes is forwarded to the final rendering stage.

Sub-process 13.5 (Display Event Board) retrieves announcements from the Announcements data store (D10) for all users. Sub-process 13.6 (Render Role-Based Dashboard View) consolidates all gathered data along with notification counts from D11 (Notifications). It then generates a personalized dashboard view based on each user's role. This ensures that users receive relevant, organized, and role-specific information upon access.

Process Flow Diagram

The Process Flow Diagram illustrates the visual overview of the STIzen system's operations, showing the step-by-step flow of processes and interactions between users, modules, and data. It highlights how tasks are executed and how information moves across the system for efficient scheduling management.

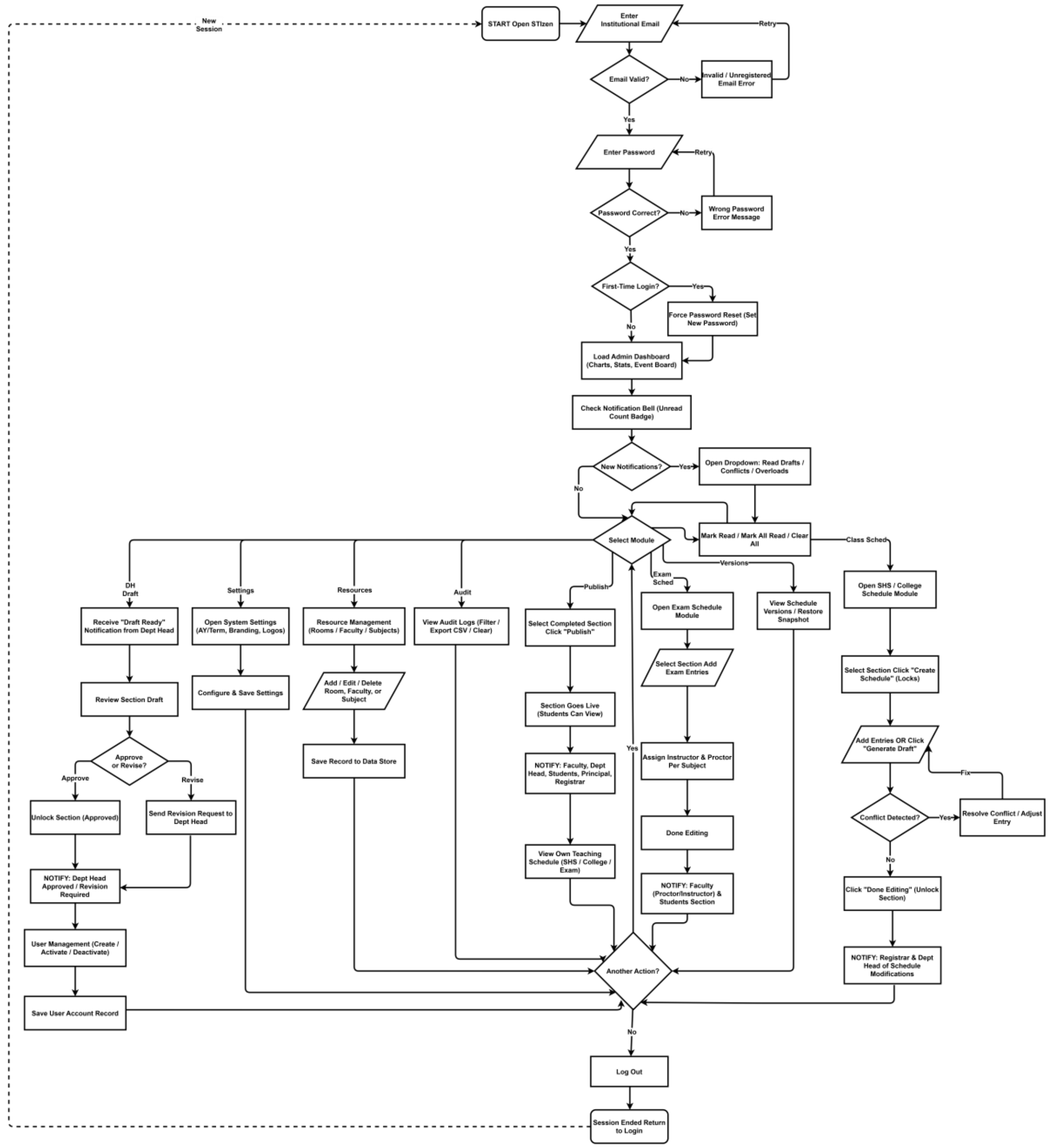

Figure 19. Academic Head Process Flow Diagram

Figure 19 above illustrates the comprehensive process flow of the Academic Head within the STIzen system. The diagram demonstrates how the Academic Head interacts with the system to perform full administrative functions, from login to system-wide schedule management and user administration. The process begins with logging in using the STI institutional email (@naga.sti.edu), where the system enforces RBAC and password security through hashing. Passwords must be a minimum of 8 characters containing at least one uppercase letter, one lowercase letter, one number, and one special character.

Upon successful authentication, the Academic Head can configure system settings, such as setting the active Academic Year (AY) and term, and customizing institutional branding including the application name and logo. The next step involves resource management, where the administrator adds or edits rooms, faculty, subjects, sections, and courses for both Senior High School (SHS) and College levels.

The Academic Head can then create, edit, and manage class schedules and exam schedules. Schedule creation can be done manually or via the "Generate Draft" feature, which automatically assigns faculty and rooms based on availability. The system performs real-time conflict detection to prevent overlapping schedules for faculty, rooms, and sections. Each schedule is version-controlled, allowing the administrator to track changes and restore previous versions if needed. Once finalized, schedules are published, triggering notifications to the Registrar, Department Heads, faculty, and students.

Additionally, the Academic Head monitors notifications for schedule drafts submitted by Department Heads, conflict alerts, and faculty overload warnings. The process flow also covers providing feedback on drafts—approving them or requesting revisions—and handling user management tasks, including creating, activating, or deactivating accounts for all roles within the system. Throughout the process, the system ensures secure, real-time updates, visibility over all academic resources, and audit logging for transparency and accountability.

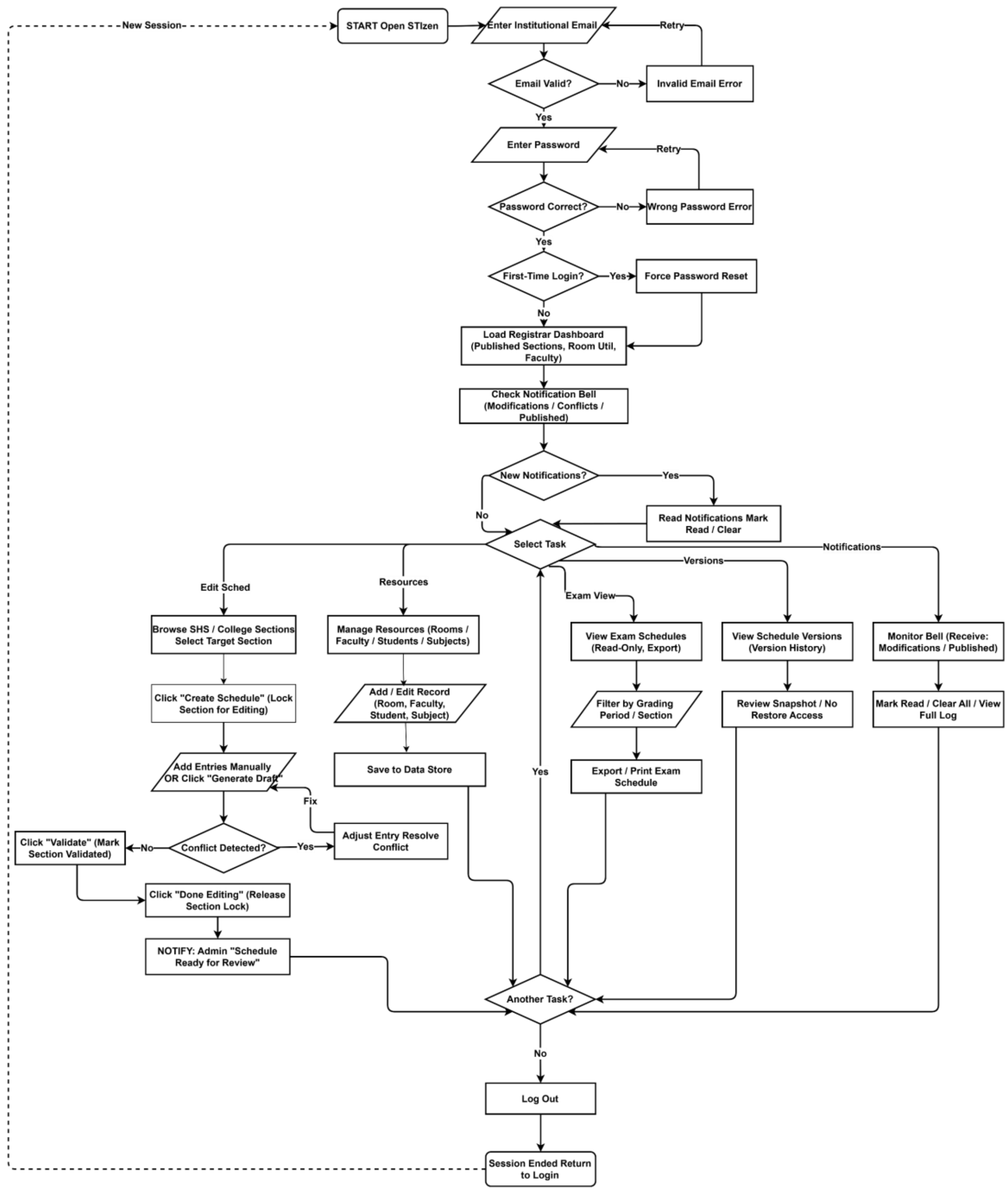

Figure 20. Registrar Process Flow Diagram

Figure 20 above illustrates the process flow of the Registrar role within the STIzen system. The diagram highlights how the Registrar interacts with the system to manage and validate class schedules while maintaining communication with administrators, department heads, faculty, and students.

The process begins with the Registrar logging in using their institutional STI email (@naga.sti.edu), where password security is enforced through hashing. Passwords must meet the minimum complexity requirement of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character. Once authenticated, the Registrar can access an admin-style dashboard to view published sections, room utilization, and active faculty.

To manage schedules, the Registrar selects a section from the SHS or College schedule list and locks it for editing, preventing concurrent modifications by other users. Schedules can be created or updated manually by adding entries for subjects, faculty, rooms, days, and times, with real-time conflict detection ensuring no overlaps occur. Alternatively, the "Generate Draft" feature allows auto-assignment of faculty and rooms according to predefined patterns, followed by a preview and adjustment of any conflicts before applying changes.

Once a schedule is complete, the Registrar validates it, marking the section as ready for review. Notifications are automatically sent to the Academic Head and other relevant users regarding schedule modifications or draft submissions. The "Done Editing" step unlocks the section and triggers alerts to administrators, ensuring that the system maintains accurate and up-to-date schedule information.

Throughout the process, the system provides read-only access to resources such as rooms, faculty, students, and subjects, enabling the Registrar to make informed scheduling decisions without compromising data integrity.

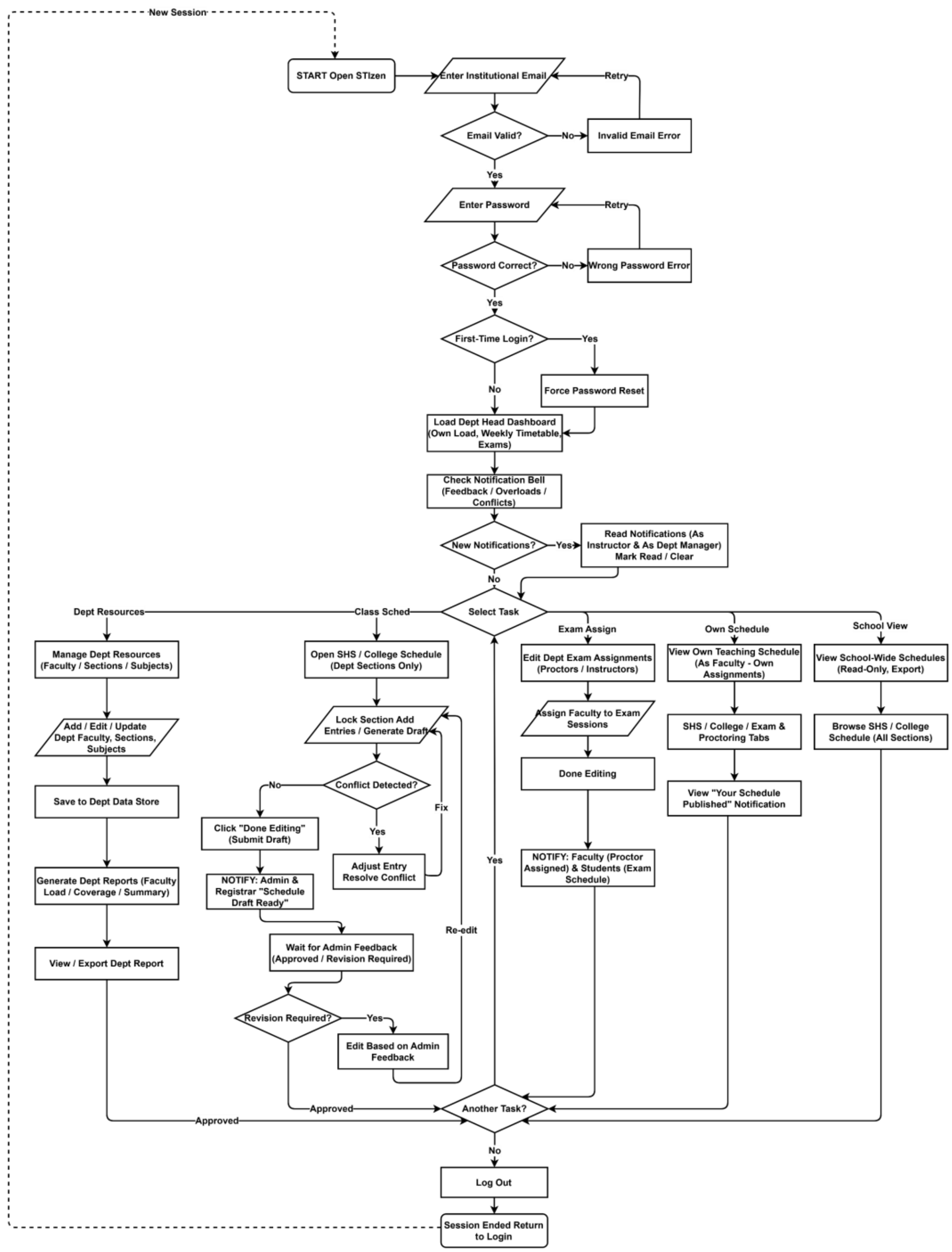

Figure 21. Department Head Process Flow Diagram

Figure 21 above illustrates the process flow of the Department Head within the STIzen system. The diagram highlights how the Department Head manages departmental schedules, faculty assignments, and academic resources while coordinating with administrators and registrars.

The process begins with the Department Head logging in using their institutional STI email (@naga.sti.edu), where RBAC and password security with hashing are enforced. Passwords must meet the minimum complexity requirement of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character. Upon authentication, the Department Head accesses a department-specific dashboard displaying load summaries, weekly timetables, upcoming exams, and notifications regarding schedule conflicts, faculty overloads, and administrative feedback.

The Department Head can manage departmental assignments by organizing faculty, sections, and subjects within their assigned department scope. For schedule management, the Department Head can lock departmental sections to create or edit SHS and College class schedules as well as exam and proctoring schedules. Schedules can be manually entered or generated as drafts using the system's auto-assignment feature, which considers faculty and room availability, with real-time conflict detection to prevent overlapping assignments.

Once schedules are finalized, notifications are sent to administrators, registrars, and assigned faculty regarding schedule changes, exam proctor assignments, or section publications. The Department Head can also generate departmental reports summarizing faculty load, section coverage, and schedule distribution. Additionally, the Department

Head can view their own teaching schedule, including SHS, College, and exam duties, and access their teaching history across academic years and terms.

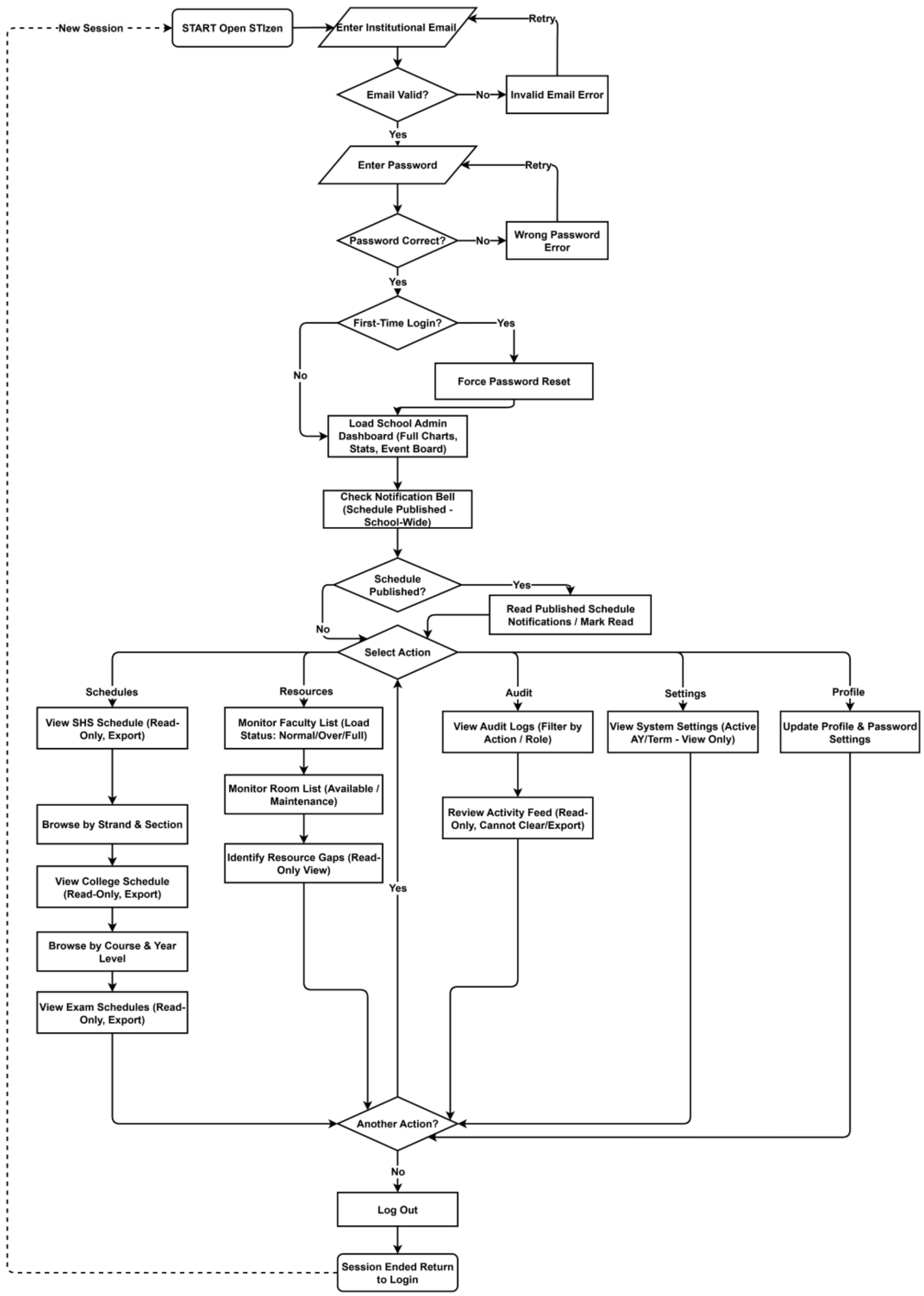

Figure 22. School Administrator Process Flow Diagram

Figure 22 above illustrates the process flow of the School Administrator within the STIzen system. The diagram highlights how the School Administrator oversees academic schedules and institutional resources through monitoring and viewing capabilities, without directly modifying scheduling data.

The process begins with the School Administrator logging in using their institutional STI email (@naga.sti.edu), where secure authentication mechanisms such as password hashing are enforced. Passwords must meet the minimum complexity requirement of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character. After successful login, the user is directed to an admin-style dashboard that presents system-wide data, including published sections, room utilization, faculty status, and event announcements. Notifications are accessed through the real-time bell icon, informing the administrator of newly published schedules across the institution.

The School Administrator can review SHS, College, and examination schedules in a read-only format, allowing them to monitor academic activities across all sections. In addition, they can view faculty and room lists to assess availability and utilization, ensuring visibility of institutional resources and operations.

Another key part of the process flow is the audit log review, where the School Administrator can filter and examine system activities performed by different users. Although they have access to these logs for transparency and monitoring, they are restricted from modifying or clearing them, preserving data integrity and accountability.

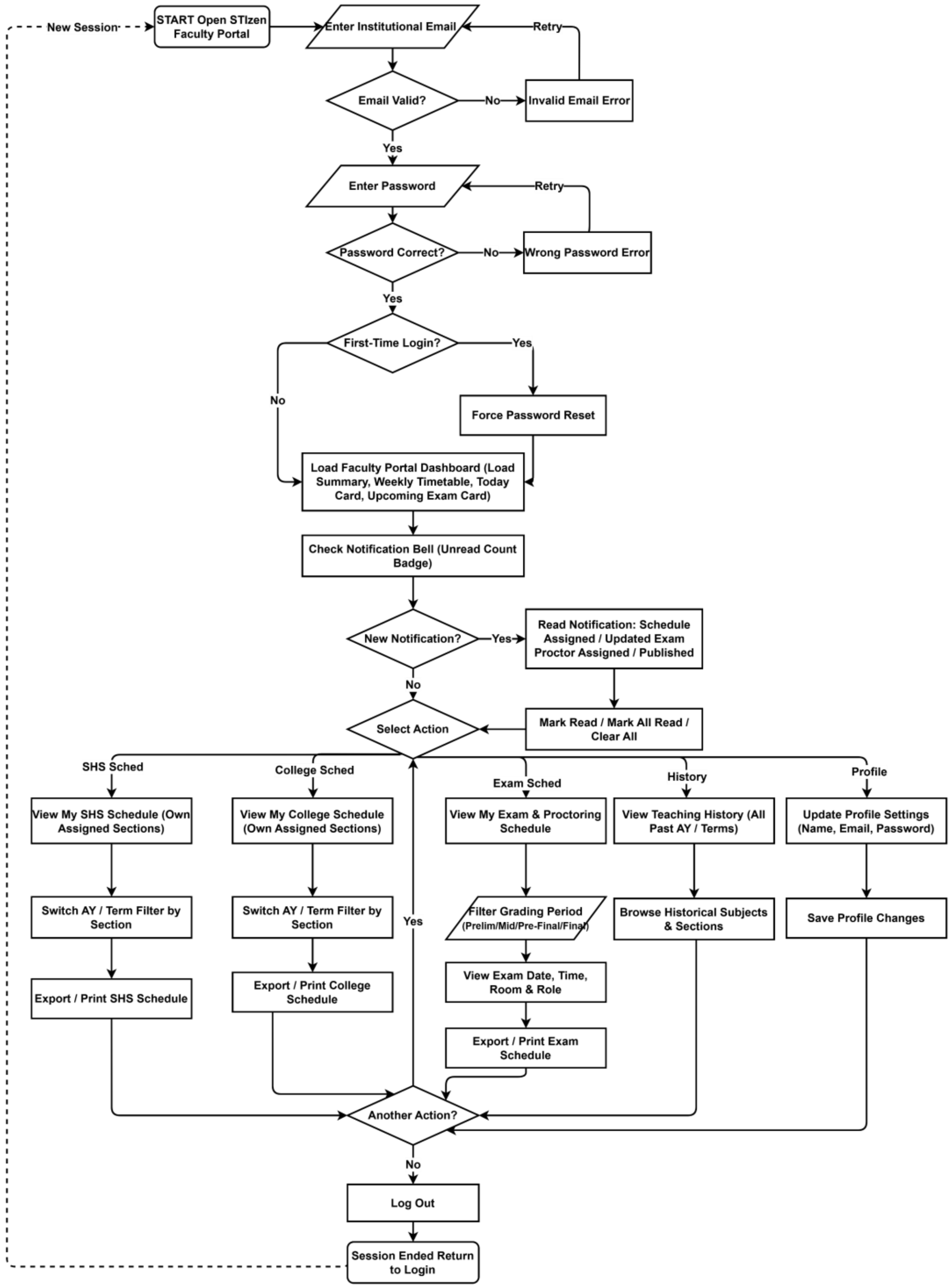

Figure 23. Faculty Process Flow Diagram

Figure 23 above illustrates the process flow of the Faculty role within the STIzen system. The diagram shows how faculty members interact with the system to access their schedules, monitor teaching loads, and stay informed about updates related to their assigned classes and responsibilities.

The process begins with the faculty member logging in using their institutional STI email (@naga.sti.edu.ph), where secure authentication is enforced through password hashing. Passwords must meet the minimum complexity requirement of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character. Upon successful login, the faculty member is directed to the faculty portal dashboard, which displays a load summary indicating assigned units, maximum load, and status, along with a weekly timetable, today’s class schedule, upcoming exams, and announcements.

Faculty members can navigate through different schedule views, including SHS schedules, College schedules, and exam or proctoring assignments. These schedules can be filtered by academic year and term, and may be exported for reference. The system also provides access to teaching history, allowing faculty to review previously handled subjects across different academic periods.

Notifications play a key role in the process, as faculty receive real-time updates through the notification bell icon. These include alerts for newly assigned schedules, updates to existing schedules, exam proctoring assignments, and confirmations when schedules are officially published. Faculty members can manage these notifications by marking them as read, archiving them, or viewing the complete notification log. Additionally, faculty members can update their personal profile information, including name, email, password, and other relevant details, through the account settings.

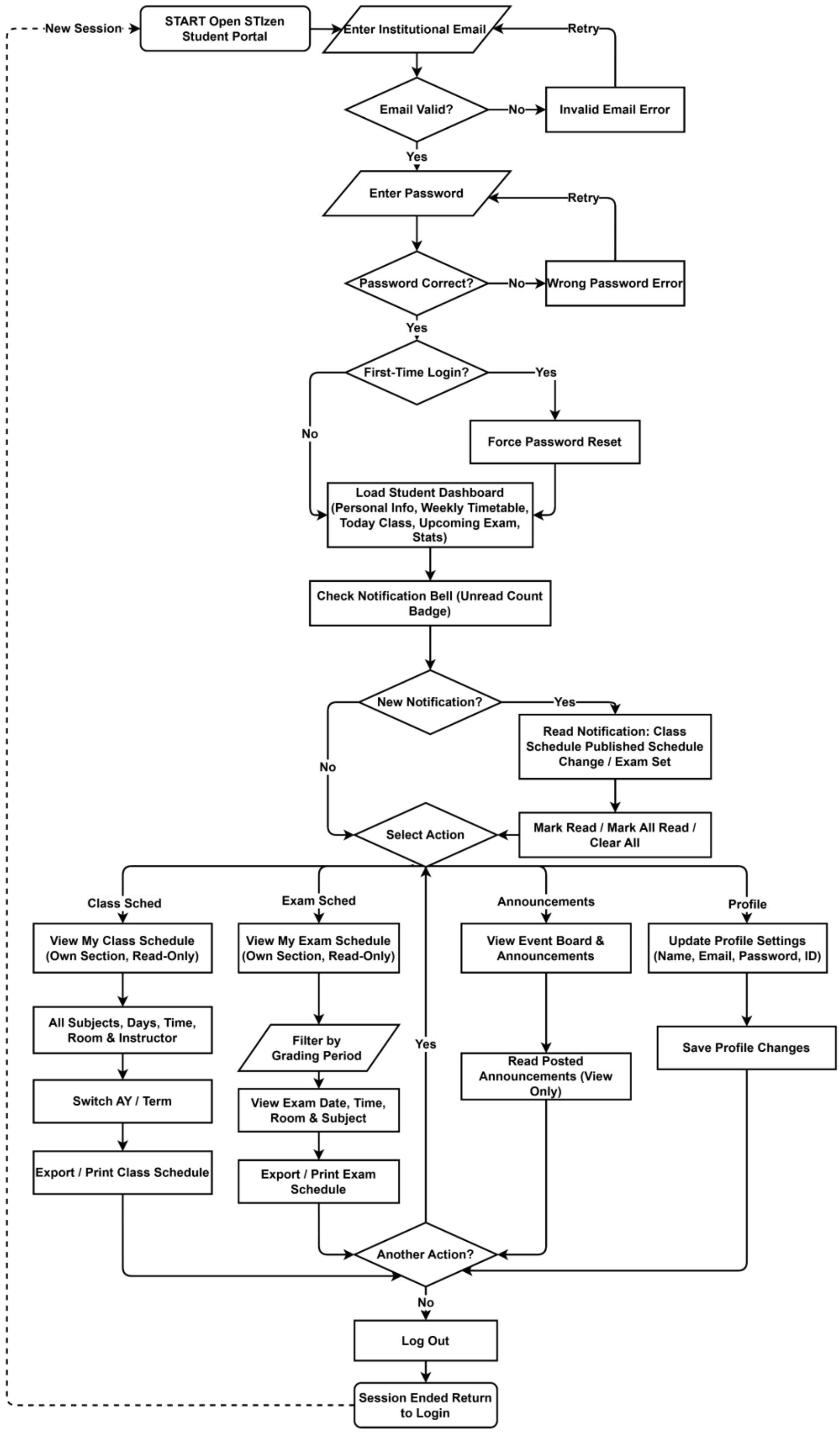

Figure 24. Student Process Flow Diagram

Figure 24 above illustrates the Student's process flow within the STIzen system. The process begins with the student logging in using their institutional STI email (@naga.sti.edu.ph), where secure password authentication ensures account safety. Passwords must meet the minimum complexity requirement of 8 characters with at least one uppercase letter, one lowercase letter, one number, and one special character to ensure secure access. Upon successful login, the dashboard presents personal details such as course, section, and current academic term, along with a weekly timetable, today's classes, and upcoming examinations for easy academic tracking. Summary statistics, including subjects and instructors, are also displayed for quick reference and better awareness of class structure.

Students can access their full class schedule, which includes subjects, assigned instructors, rooms, and time slots for each class. They also have the option to switch between academic years and terms, allowing them to view past and current schedules, and can export or print their schedules for offline reference. Examination schedules are viewable by grading period, enabling students to track exam dates, times, and assigned rooms in an organized manner.

Notifications keep students informed of system updates, such as new or updated class schedules and examination postings that may affect their classes. Alerts are delivered in real time through the dashboard bell icon and can be marked as read, archived, or reviewed in full history for later reference. Finally, students can manage their account settings by updating personal information such as name, email, student ID, address, birthday, and password to ensure their profile remains accurate and up to date.

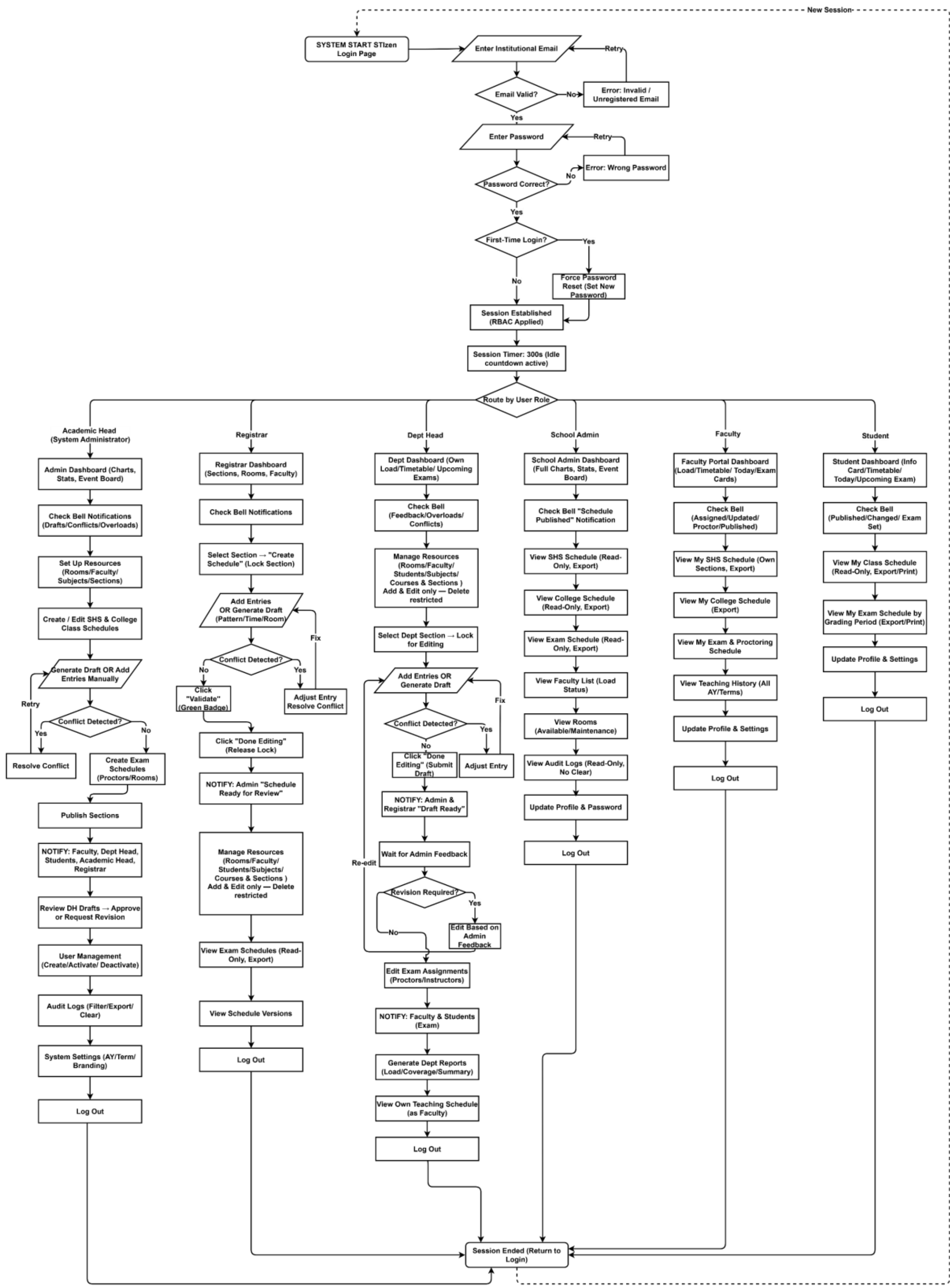

Figure 25. Combined System Process Flow Diagram

Figure 25 above presents the unified process flow diagram of the STIzen Academic Scheduling Management System, illustrating the complete interaction of all six user roles, the Academic Head, Registrar, Department Head, School Administrator, Faculty, and Student within a single consolidated view. The diagram begins with a shared login flow where all users enter their institutional STI email and password, passing through email validation, password verification, and a first-time login check that triggers a forced password reset when applicable. Upon successful authentication, the system establishes a role-based session with a 300-second idle timeout and routes each user to their designated dashboard based on their assigned role.

The Academic Head, as the system's super-user, has access to the full range of administrative functions including resource management, class and exam schedule creation, conflict detection and resolution, section locking and publishing, draft review and approval, user account management, audit log monitoring, and system configuration. As the primary scheduler of the institution, the Academic Head also oversees the auto-suggestion engine during conflict resolution, maintains full control over schedule versioning, and serves as the final approving authority for all schedule drafts submitted by Department Heads before they are published institution-wide.

The Registrar supports scheduling operations by managing class schedules and resources including rooms, faculty, students, subjects, and courses and sections with add and edit permissions only, and views exam schedules in a read-only format. The Registrar plays a critical supporting role in enrollment management by ensuring that section records, room availability, and subject assignments remain accurate and up to date throughout the academic term, and receives real-time notifications whenever schedules are published or modified by the Academic Head.

The Department Head operates within the scope of their assigned department, managing departmental resources, drafting class and exam schedules, editing proctor assignments, and generating departmental reports, with add and edit permissions only and no delete access to master data records. After completing a schedule draft, the Department Head submits it for review through the system, waits for administrative feedback, and makes revisions as required before the schedule is finalized and approved by the Academic Head, ensuring that departmental scheduling aligns with institution-wide academic policies.

The School Administrator serves a monitoring and oversight role, with read-only access to all published SHS and College class schedules, exam schedules, faculty and room records, audit logs, and system settings, without the ability to modify, create, or delete any data within the system. This role is designed to provide institutional leadership with a transparent, real-time view of academic operations across all departments, enabling informed decision-making based on accurate and up-to-date scheduling information without interfering with the scheduling workflow.

Faculty members access only their own assigned teaching and proctoring schedules, teaching history, and bell notifications relevant to their duties. Students have read-only access to their section's class and exam schedules, real-time notifications regarding schedule updates, and personal account settings. All roles share a common session termination step that returns the user to the login page upon logout or session expiry.

Use Case Diagram

The Use Case Diagram illustrates the interactions between users and the system by identifying different user roles such as the Academic Head, Registrar, Department Head, School Administrator, Faculty, and Student, and showing how each role interacts with specific system features and functionalities. Separate diagrams are provided for each role to clearly present their respective actions and responsibilities, reduce visual complexity, and define access boundaries. Overall, the diagrams provide a structured overview of system behavior and user responsibilities to support clear understanding and implementation.

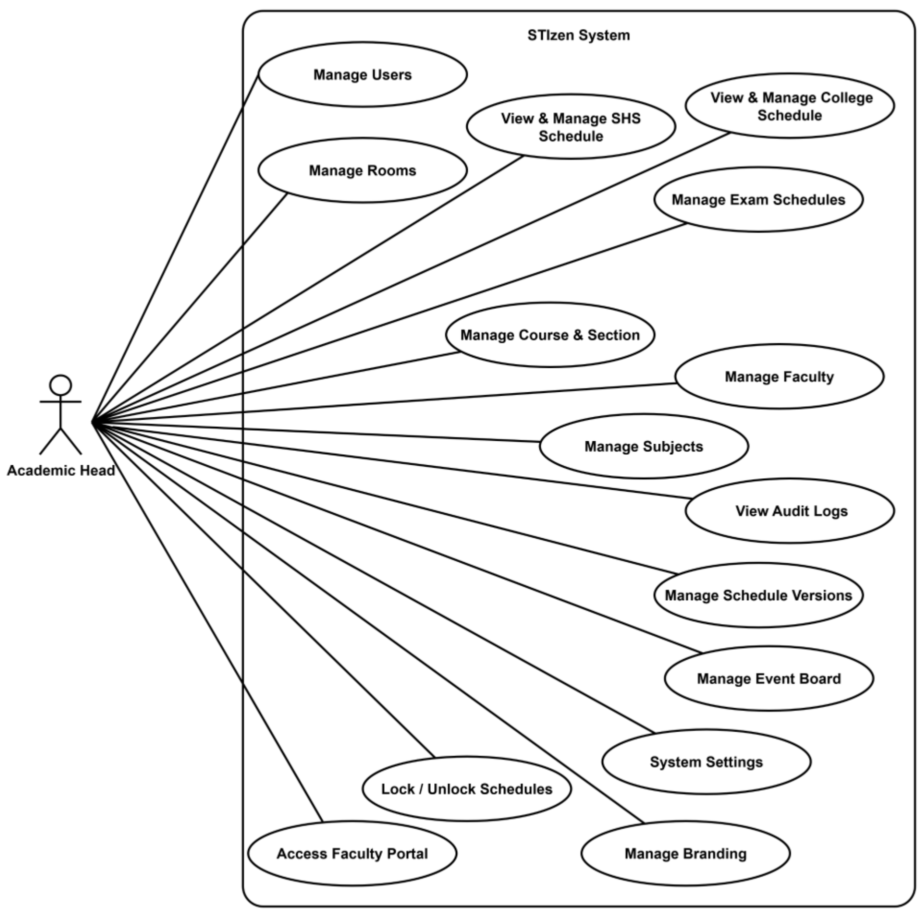

Figure 26. Academic Head Use Case Diagram

Figure 26 above illustrates the use case diagram of the Academic Head within the system. The Academic Head serves as the super-user with full access to all system modules across the institution. The diagram shows responsibilities including managing user accounts for all roles, creating and editing schedules for SHS and College programs, and overseeing exam scheduling. It also includes management of master data such as rooms, faculty, courses and sections, and subjects. Additionally, the Academic Head monitors system activity through audit logs, manages schedule versions, posts announcements via the Event Board, and configures system settings and branding. This figure highlights the Academic Head's central role in ensuring the smooth operation, integrity, and accessibility of the STIzen platform.

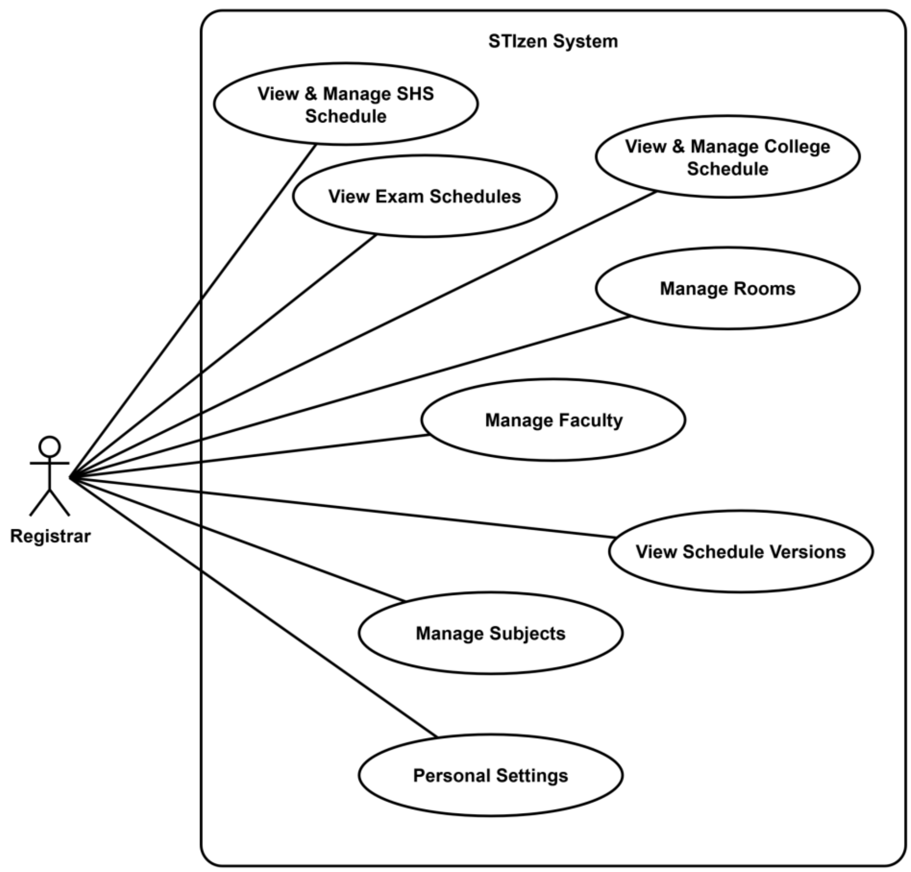

Figure 27. Registrar Use Case Diagram

Figure 27 above illustrates the use case diagram of the Registrar within the system. The Registrar is responsible for academic record management and scheduling workflows. The diagram highlights functions such as creating, editing, and managing class schedules for both SHS and College programs, viewing exam schedules, and maintaining master data including rooms, faculty, courses and sections, and subjects. The Registrar is permitted to add and edit master data entries but does not have permission to delete records, ensuring controlled and secure data management within the system.

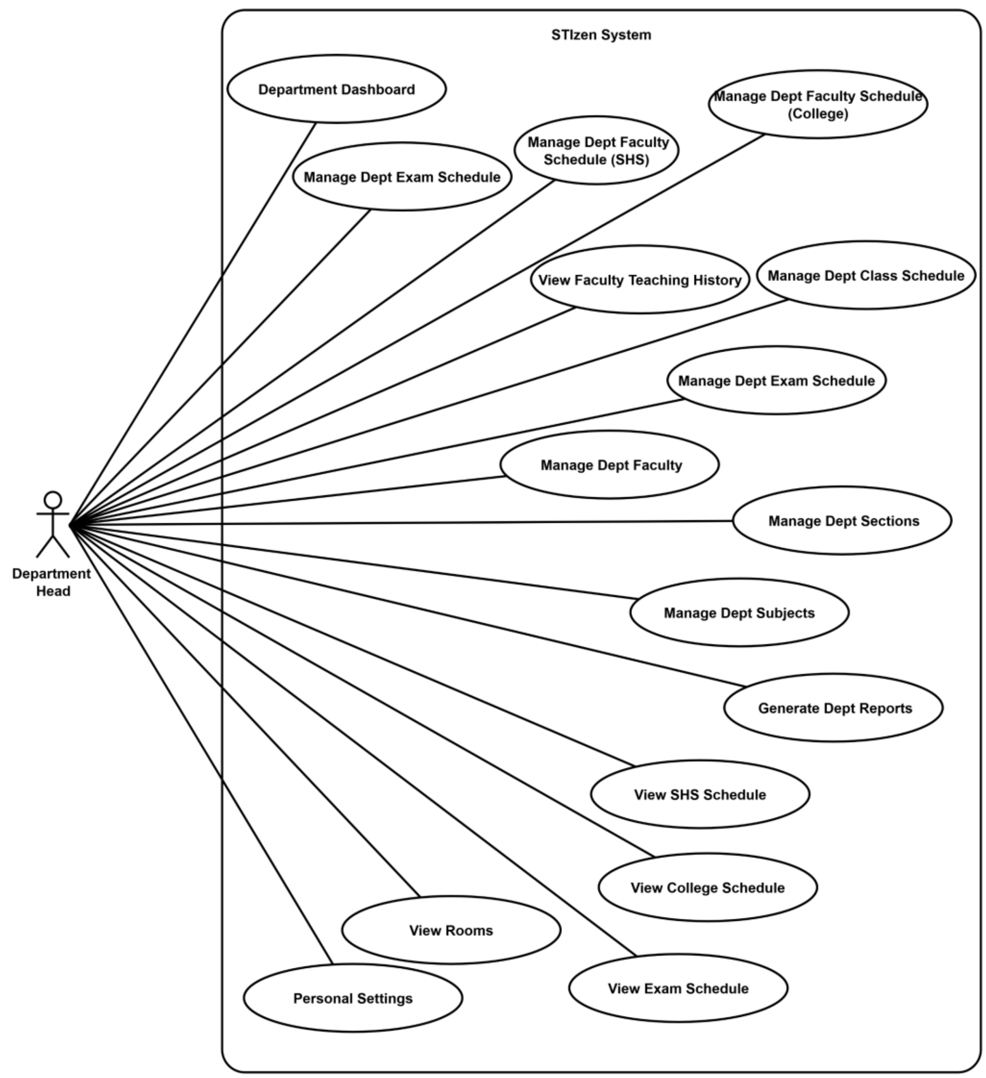

Figure 28. Department Head Use Case Diagram

Figure 28 above illustrates the use case diagram of the Department Head within the system, who is responsible for managing academic scheduling, faculty assignments, sections, and subjects within their assigned department. The diagram highlights the Department Head's ability to draft SHS and College schedules, coordinate faculty exam schedules, and organize departmental sections and subjects. It also includes the generation of departmental reports on faculty workload and section coverage. In addition, the Department Head has read-only access to schedules to support cross-department coordination. This figure emphasizes a draft-and-submit workflow, where schedules are submitted to the Academic Head for review and approval.

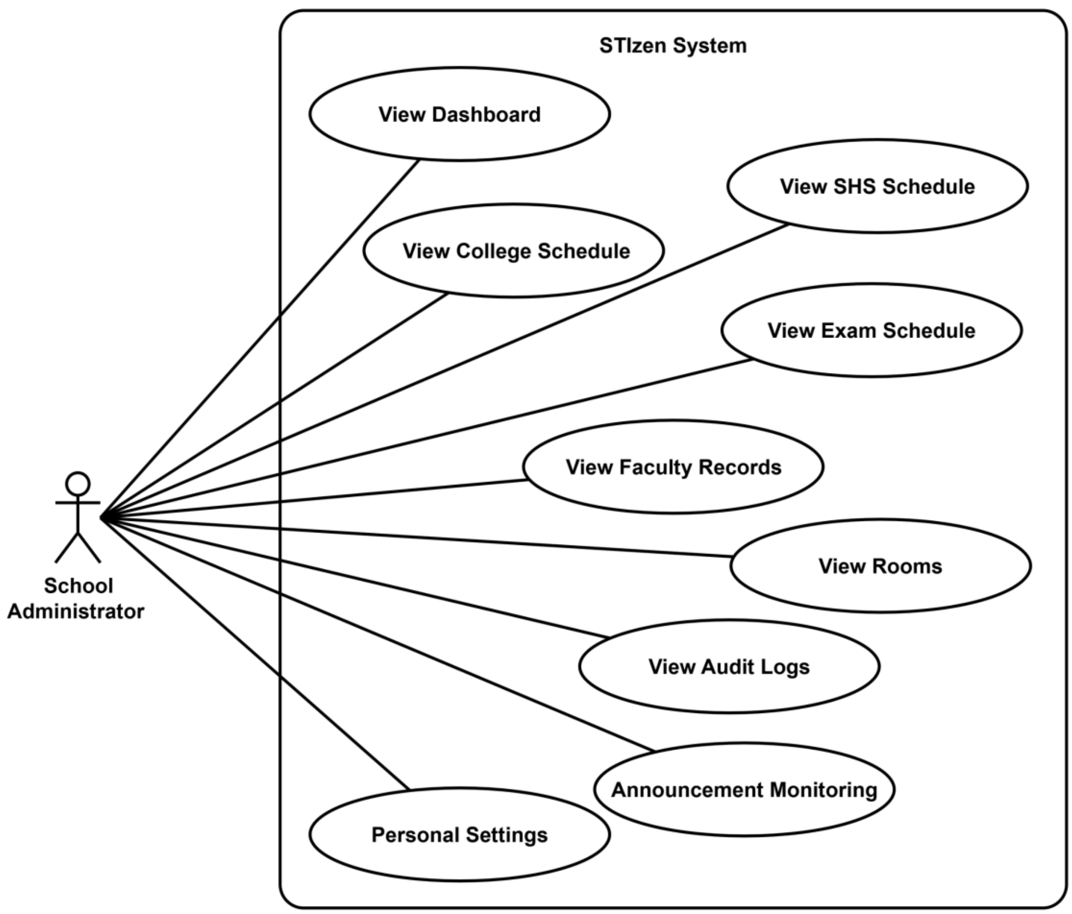

Figure 29. School Administrator Use Case Diagram

Figure 29 above illustrates the use case diagram of the School Administrator, who supervises institution-wide scheduling and events primarily for monitoring and reporting purposes. The diagram shows how the School Administrator can review SHS and College schedules, oversee exam schedules, monitor departmental performance, and monitor announcements. Their access is largely read-only with supervisory privileges, ensuring smooth operation without direct data modification. This figure highlights the role's focus on transparency and adherence to academic policies.

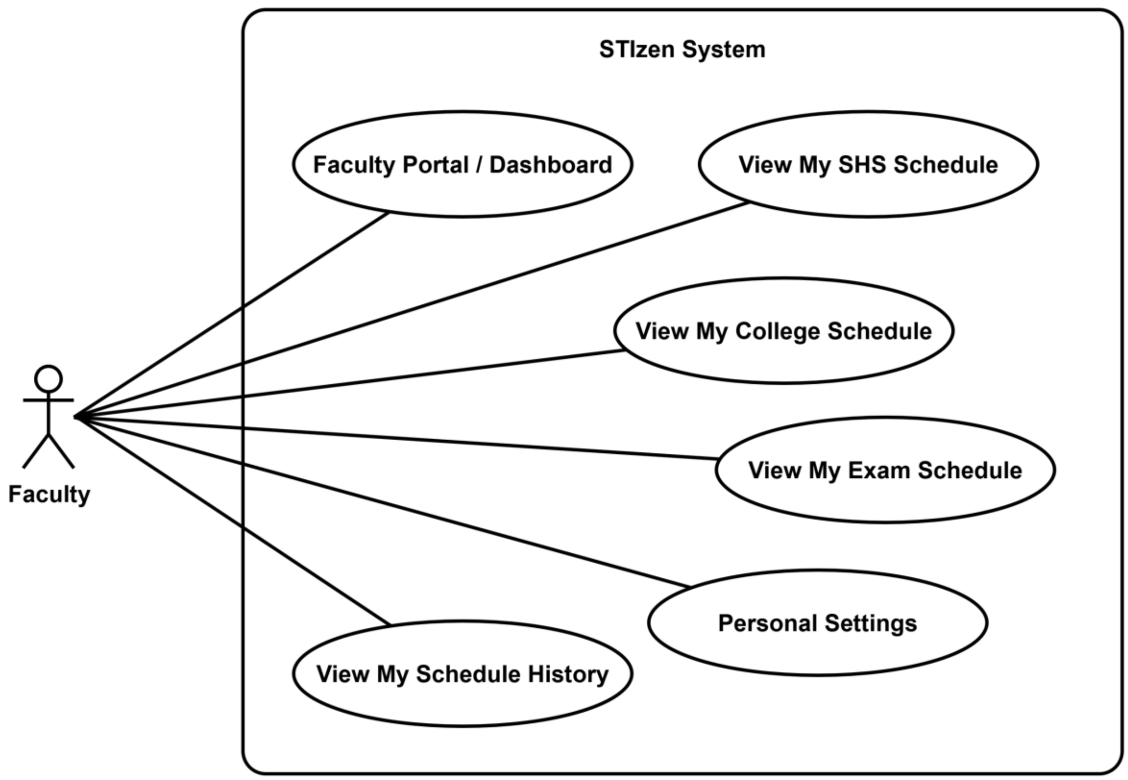

Figure 30. Faculty Use Case Diagram

Figure 30 above illustrates the use case diagram of the Faculty, who primarily interact with STIzen to view their teaching assignments and schedules. The diagram highlights responsibilities such as accessing their assigned SHS and College class schedules, viewing examination and proctoring assignments, monitoring their teaching history, and managing personal settings through the faculty portal dashboard. Faculty also receive notifications about schedule changes and relevant events. This figure emphasizes faculty participation in operational tasks while maintaining limited access to sensitive system data.

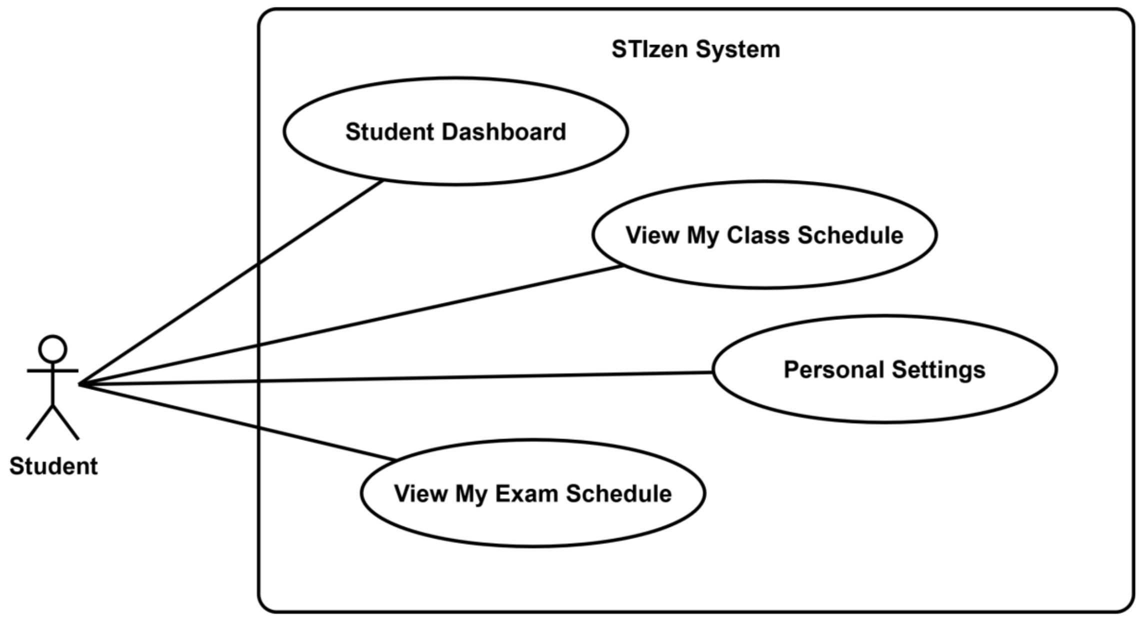

Figure 31. Student Use Case Diagram

Figure 31 above illustrates the use case diagram of the student, who uses STIzen mainly to access class schedules, exam schedules, announcements, and notifications. The diagram shows the student's read-only access to schedules for their enrolled strand or course, enabling them to track academic activities and stay informed. While students cannot modify schedules or system data, they receive updates and notifications relevant to their academic progress. This figure highlights the system's role in facilitating communication and information access for students.

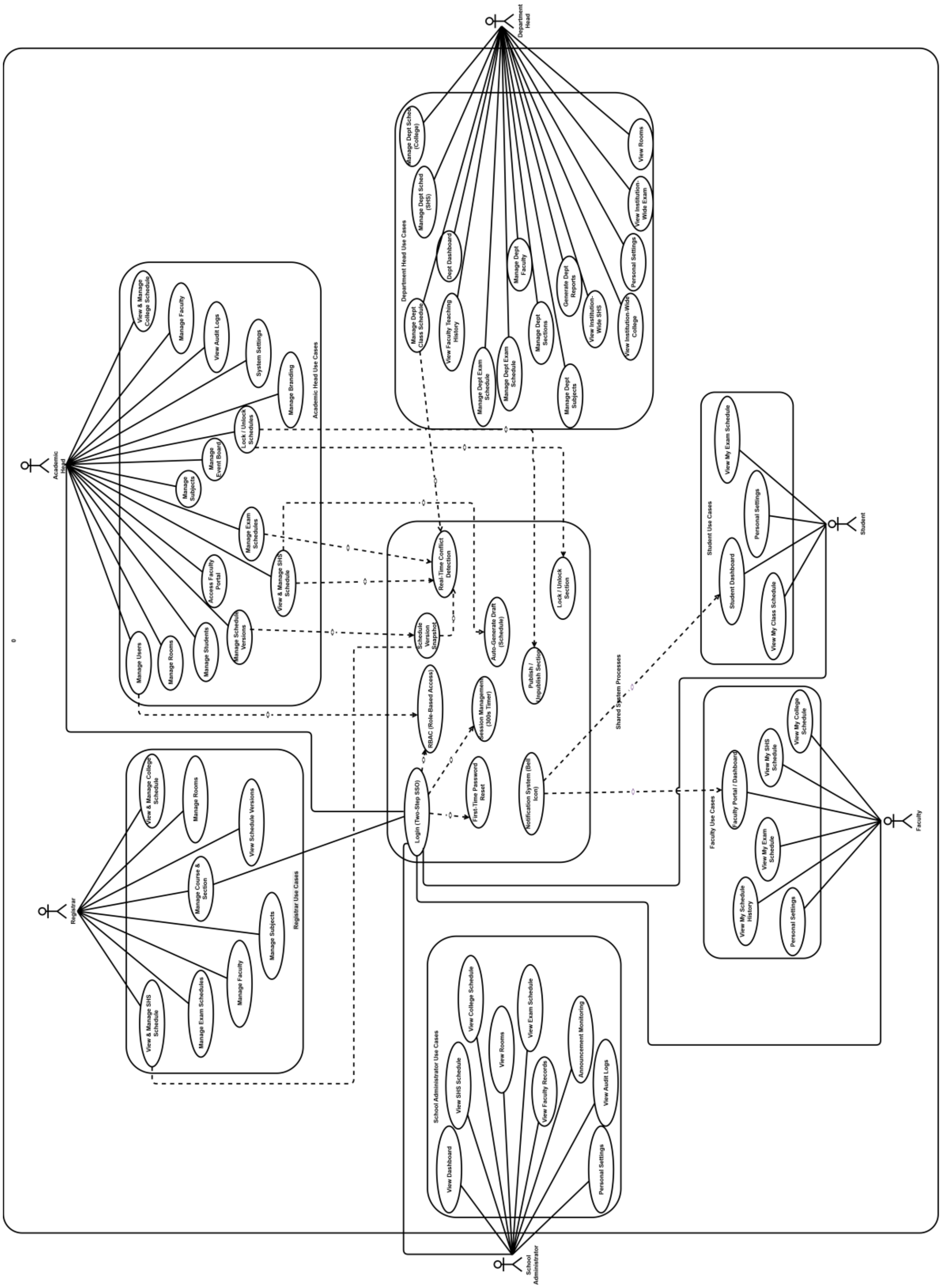

Figure 32. Combined Use Case Diagram

Figure 32 above presents the comprehensive use case diagram of the STIzen Academic Scheduling Management System, illustrating the interactions between six primary actors—Academic Head, Registrar, Department Head, School Administrator, Faculty, and Student—and their respective system functionalities. The diagram is organized into actor-specific boundaries and a centralized Shared System Processes boundary that includes common system functions.

The Academic Head has full access to all modules, including user management, SHS and College scheduling, resource management, schedule versions, audit logs, event board, and system settings. The Registrar manages class schedules and maintains master data with add and edit permissions only. The Department Head handles department-level scheduling, resources, and reports, with read-only access to institution-wide data. The School Administrator has read-only access for monitoring schedules, resources, audit logs, and announcements. Faculty members access assigned schedules, exam duties, teaching history, and personal settings, while Students have read-only access to their schedules and account information.

The Shared System Processes boundary contains functionalities shared across multiple roles, including Login with Two-Step SSO, Role-Based Access Control (RBAC), Schedule Version Snapshot, Real-Time Conflict Detection, First-Time Password Reset, Session Management, Notification System, Auto-Generate Draft Schedule, Publish/Unpublish Section, and Lock/Unlock Section. These shared processes form the security, automation, and communication backbone of the system, ensuring that all role-specific actions are governed by consistent access controls and real-time validation mechanisms.

Test Case

This section presents the planned testing of the system to verify the correct functionality of its features across different user roles. Test cases were designed based on the system's use cases, focusing on critical operations such as login, user management, schedule creation, conflict detection, schedule publishing, and notification handling.

System interface illustrations and descriptions show key test cases to demonstrate expected system behavior and user interactions. A comprehensive test case table follows, listing all scenarios, test steps, expected results, and status for prototype testing. These test cases ensure that all user roles and functionalities are properly verified. This plan helps identify and fix issues early, ensuring the system works correctly for all users.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-A01	Valid Admin login	1. Open app 2. Click Sign in with STI Email 3. Enter email & password 4. Click Login	admin@naga.sti.edu Pass@2024	Admin Dashboard opens; full sidebar navigation visible
TC-A02	Invalid email domain	1. Enter non-STI email 2. Click Continue	user@gmail.com	Error: 'Please use your STI institutional email (@naga.sti.edu.ph)'
TC-A03	Unregistered email	1. Enter valid domain but unknown email 2. Click Continue	ghost@naga.sti.edu	Error: 'Email not registered. Contact your administrator.'
TC-A04	Empty email field	1. Leave email blank 2. Click Continue	(blank)	Error: 'Please enter your STI email.'
TC-A05	Wrong password	1. Enter valid admin email 2. Enter wrong password 3. Click Login	PW: wrongpass	Error message shown; login denied
TC-A06	Password visibility toggle	1. Enter password 2. Click eye icon 3. Click again	Pass@2024	Password toggled between visible text and hidden dots
USER MANAGEMENT				
TC-A07	View all users	1. Go to User Management	Role: admin	All users listed with name, role, email, dept, status
TC-A08	Add new user – valid	1. Click Add User 2. Fill all fields 3. Save	Name: New User Role: faculty Email: new@naga.sti.edu PW: Test@2024	User added; success toast; appears in list

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-A09	Add user – duplicate email	1. Add user with existing email	admin@naga.sti.edu	Error: 'Email already in use'
TC-A10	Add user – invalid email domain	1. Add user with non-STI email	test@gmail.com	Error: email must be @naga.sti.edu.ph
TC-A11	Edit existing user	1. Click Edit on user 2. Update name/dept 3. Save	Change any field	Changes saved; audit log entry created
TC-A12	Deactivate user	1. Toggle user active status to inactive	Any active user	User deactivated; cannot login while inactive
TC-A13	Delete user	1. Click Delete on user 2. Confirm dialog	Any non-admin user	User removed after confirmation
TC-A14	Filter users by role tab	1. Click Faculty / Student / System Users tabs	Role filter tabs	Only matching-role users displayed
PASSWORD POLICY				
TC-A15	Password too short	1. Change password 2. Enter < 8 character password	PW: Ab1!	Error: 'Password must be at least 8 characters long.'
TC-A16	Missing uppercase	1. Enter password without uppercase	PW: pass@2024	Error: 'Password must contain at least one uppercase letter.'
TC-A17	Missing lowercase	1. Enter password without lowercase	PW: PASS@2024	Error: 'Password must contain at least one lowercase letter.'
TC-A18	Missing digit	1. Enter password without number	PW: Password@	Error: 'Password must contain at least one number.'
TC-A19	Missing special character	1. Enter password without special char	PW: Password1	Error: 'Password must contain at least one special character.'
TC-A20	Valid strong password	1. Enter password meeting all rules 2. Save	PW: StrongPass@2024	Password accepted and saved successfully
SHS CLASS SCHEDULE				
TC-A21	View SHS schedule by section	1. Go to Senior High School 2. Select strand & section	Section: ABM-11A	Schedule table shows all entries: subject, faculty, room, day, time
TC-A22	Add SHS schedule entry – valid	1. Click Add Entry 2. Fill all fields 3. Save	Subject: English Faculty: Ana Dela Cruz Room: RM101 Day: MWF Time: 8:00–9:00 AM	Entry added; version snapshot captured
TC-A23	Conflict – same room same time	1. Add entry with already-booked room & time	Room: RM101 at 8:00 AM (booked)	Conflict alert; entry not saved
TC-A24	Conflict – faculty double booking	1. Add entry with faculty already scheduled at same slot	Faculty booked at 8:00 AM Monday	Conflict alert: faculty schedule clash
TC-A25	Edit SHS schedule entry	1. Click Edit 2. Modify room or time 3. Save	Change room RM101 → RM102	Entry updated; conflict recheck runs; version saved
TC-A26	Delete SHS schedule entry	1. Click Delete on entry 2. Confirm	Any existing entry	Entry removed immediately

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-A27	Lock section schedule	1. Click Lock on section	Section: ABM-11A	Section locked; edits disabled while locked
TC-A28	Publish section schedule	1. Click Publish on section	Section: ABM-11A	Section published; visible to faculty and students
TC-A29	Generate Draft schedule	1. Click Generate Draft 2. Set pattern, time, duration 3. Preview 4. Apply	Pattern: MWF Start: 8:00 AM Duration: 1.5 hrs	Draft auto-generated with no conflicts; applied on confirm
COLLEGE (TERTIARY) CLASS SCHEDULE				
TC-A30	View college schedule by section	1. Go to Tertiary 2. Select course & section	Section: BSIT-1A	Schedule displayed with all entries
TC-A31	Add college entry – valid	1. Click Add Entry 2. Fill all fields 3. Save	Subject: Programming Faculty: Dela Cruz Room: GRF01 Day: TTH	Entry saved; version captured
TC-A32	Conflict – computer lab double-booked	1. Book Comlab 1 at same slot twice	Room: GRF01 at 10:00 AM Thursday	Conflict alert; save blocked
TC-A33	Merged section display	1. View schedule for merged year level	Section: BSIT-3 (merged)	Merged section note displayed correctly
EXAM SCHEDULE				
TC-A34	View SHS exam schedule	1. Go to Exam Schedule > SHS tab 2. Select section	Section: ABM-11A / Midterm	Exam table shows subject, proctor, room, date, time
TC-A35	View College exam schedule	1. Go to Exam Schedule > College tab 2. Select section	Section: BSIT-1A / Prelim	College exam schedule displayed
TC-A36	Add exam entry – valid	1. Click Add Exam Entry 2. Fill all fields 3. Save	Subject, room, proctor, date, time	Exam entry added; version captured
TC-A37	Exam conflict – room double-booked	1. Add exam with room/time already taken	Same room, overlapping time	Conflict alert; entry not saved
TC-A38	Add special exam entry	1. Click Add Exam Entry 2. Select "Special Exam" as type 3. Fill all fields 4. Save	Exam type: Special, Subject, room, proctor, date, time	Special exam entry added; labeled as Special Exam; version captured
TC-A39	Special exam conflict detection	1. Add special exam entry with room/time already taken by another exam	Same room, overlapping time	Conflict alert; entry not saved
ROOM MANAGEMENT				
TC-A40	View all rooms	1. Go to Room Availability	Role: admin	All rooms listed with type, capacity, status
TC-A41	Add new room	1. Click Add Room 2. Fill all fields 3. Save	Name: Room 401 Type: Regular Capacity: 40 Status: Available	Room added to list

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-A42	Edit room – set to Maintenance	1. Click Edit on room 2. Change status 3. Save	Room 307 → Maintenance	Status updated; room excluded from scheduling dropdowns
TC-A43	Maintenance room excluded from scheduling	1. Open schedule modal 2. Check room dropdown	Room 307 (Maintenance)	Maintenance room not selectable
TC-A44	Delete room	1. Click Delete on room 2. Confirm	Empty room with no bookings	Room removed from list
SCHEDULE VERSIONS				
TC-A45	Version auto-captured on schedule change	1. Add/edit/delete any schedule entry	Any schedule modification	New version created with timestamp, user, description
TC-A46	View version history	1. Go to Schedule Versions	Role: admin	Up to 50 most recent versions listed
TC-A47	Restore previous version	1. Click Restore on a version 2. Confirm	Any past version	Schedule reverted; current state auto-snapshotted before restore
TC-A48	Delete a version	1. Click Delete on version 2. Confirm	Any existing version	Version removed permanently
TC-A49	Compare versions (diff view)	1. Click View on a version	Any version	Diff panel shows added/removed entries vs previous
AUDIT LOGS				
TC-A50	Audit log created on login	1. Login as Admin	Valid credentials	Log entry created: user, role, timestamp, action
TC-A51	Audit log on schedule changes	1. Add/edit/delete a schedule entry	Any schedule change	Log records what changed and who changed it
TC-A52	Audit log on user management	1. Add/edit/delete a user	User management actions	Each action logged with type and actor
TC-A53	View full audit log	1. Go to Audit Logs	Role: admin	Full log list displayed with filters
EVENT BOARD				
TC-A54	View event board	1. Go to Event Board	Role: admin	All event posts listed with title, description, image, posted by/at
TC-A55	Add new event post	1. Click Add Post 2. Fill title, description 3. Save	Title: Enrollment Schedule Description: …	Post added and displayed on board
TC-A56	Delete event post	1. Click Delete on post 2. Confirm	Any existing post	Post removed from board

TC #	Test Scenario	Test Steps	Test Data	Expected Result
NOTIFICATIONS				
TC-A57	Notification triggered on schedule publish	1. Publish a section schedule	Published section	Notification sent to assigned faculty and students of that section
TC-A58	Mark notification as read	1. Click on unread notification	Unread notification	Notification marked as read; unread count badge decreases
SETTINGS				
TC-A59	Change active AY and Term	1. Go to Settings > System 2. Change AY/Term 3. Save	AY: 2025–2026 / 1st Semester	Dashboard and schedules reflect new AY/Term; audit log created
TC-A60	Update security email	1. Go to Settings > Security 2. Change email 3. Save	updated@naga.sti.edu.ph	Email updated; must be @naga.sti.edu.ph domain
TC-A61	Security email duplicate check	1. Enter email of another existing user	m.santos@naga.sti.edu.ph	Error: 'Email already in use by another user'
TC-A62	Upload app logo (branding)	1. Settings > Branding 2. Upload logo 3. Save Branding	Valid PNG/JPG	Logo shown in sidebar and login navbar
TC-A63	Upload school logo	1. Upload school logo 2. Save Branding	Valid PNG/JPG	School logo shown on login card
TC-A64	Change app name	1. Edit App Name 2. Save Branding	App Name: STIzen Custom	Browser tab title and sidebar brand updated
TC-A65	Reset all data	1. Click Reset All Data 2. Confirm	All system data	All data cleared; page reloads to defaults

Table 5. Test Cases for Academic Head

Table 5 above presents the test cases for the Academic Head role, which has full system access. Login test cases (TC-A01 to TC-A06) verify that valid credentials grant access to the admin dashboard while invalid inputs are properly rejected. User Management (TC-A07 to TC-A14) and Password Policy (TC-A15 to TC-A20) ensure proper execution of user creation, updating, deletion, duplicate email detection, and enforcement of password complexity requirements. Schedule test cases for SHS (TC-A21 to TC-A29), College (TC-A30 to TC-A33), and Exams (TC-A34 to TC-A39) validate schedule creation, editing, conflict detection, locking, and publishing functions.

Room Management (TC-A40 to TC-A44), Schedule Versions (TC-A45 to TC-A49), and Audit Logs (TC-A50 to TC-A53) verify room status control, version tracking with restore and comparison features, and system activity logging. Event Board (TC-A54 to TC-A56), Notifications (TC-A57 to TC-A58), and Settings (TC-A59 to TC-A65) test announcement management, real-time notification handling, and system configuration features such as academic year and term settings, email security, branding, and full data reset. These tests ensure that the Academic Head can effectively manage all system operations while maintaining data integrity, security, and overall system performance.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-R01	Valid Registrar login	1. Open app 2. Enter email & password 3. Click Login	santos@naga.sti.edu Pass@2024	Registrar Dashboard opens; nav show
TC-R02	Invalid email / wrong password	1. Enter wrong credentials	Wrong email or password	Error shown; access denied
SHS CLASS SCHEDULE				
TC-R03	View SHS schedule by section	1. Go to Senior High School 2. Select section	Section: ABM-11A	Schedule table with all entries shown
TC-R04	Add SHS schedule entry – valid	1. Click Add Entry 2. Fill all fields 3. Save	Subject, faculty, room, day, time	Entry added; version snapshot captured
TC-R05	Conflict – room double-booked	1. Add entry with already-booked room/time	Same room at same time	Conflict alert; entry not saved
TC-R06	Conflict – faculty double booking	1. Assign faculty already scheduled same slot	Faculty booked at same time	Conflict alert shown
TC-R07	Edit SHS entry	1. Edit entry 2. Change room/time 3. Save	Any field change	Entry updated; version saved
TC-R08	Delete SHS entry	1. Delete entry 2. Confirm	Any entry	Entry removed
TC-R09	Lock section	1. Lock a section	Section: STEM-11A	Section locked; edits blocked
TC-R10	Publish section	1. Publish a section	Section: STEM-11A	Section published; visible to students and faculty
TC-R11	Generate Draft schedule	1. Click Generate Draft 2. Configure 3. Apply	Pattern, start time, duration	Draft auto-generated and applied
COLLEGE (TERTIARY) CLASS SCHEDULE				
TC-R12	View college schedule	1. Go to Tertiary 2. Select course & section	Section: BSHM-2A	Schedule displayed

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-R13	Add college entry – valid	1. Add entry with valid data	No conflict data	Entry saved
TC-R14	Conflict detection	1. Add entry with duplicate room/time	Booked slot	Conflict alert; save blocked
EXAM SCHEDULE				
TC-R15	View SHS exam schedule	1. Go to Exam > SHS tab 2. Select section	Section: ABM-11A / Midterm	Exam entries shown
TC-R16	View College exam schedule	1. Exam > College tab 2. Select section	Section: BSIT-1A / Prelim	College exam entries shown
TC-R17	View exam data	1. View exam data 2. Save	Valid exam data	Exam data viewed
TC-R18	Exam room conflict	1. Add exam entry on booked slot	Same room/time	Conflict alert; not saved
FACULTY MANAGEMENT				
TC-R19	View faculty list	1. Go to Faculty	Role: registrar	All faculty shown with units info
TC-R20	Add new faculty	1. Click Add Faculty 2. Fill fields 3. Save	Name, dept, max units	Faculty added to list
TC-R21	Edit faculty	1. Edit faculty 2. Change dept 3. Save	Any field change	Changes saved
ROOM MANAGEMENT				
TC-R22	View all rooms	1. Go to Room Availability	Role: registrar	Rooms listed with type, capacity, status
TC-R23	Room status affects scheduling dropdown	1. Open schedule modal 2. Check rooms	Maintenance room present	Maintenance rooms excluded from dropdown
TC-R24	Delete room blocked	1. Attempt to delete a room	Any room	No delete button visible; action denied
SUBJECT MANAGEMENT				
TC-R25	Add new subject	1. Go to Subjects 2. Click Add Subject 3. Fill fields 4. Save	Subject name, strand/course	Subject added to list
TC-R26	Edit existing subject	1. Click Edit on subject 2. Update field 3. Save	Any subject	Changes saved
TC-R27	Delete subject blocked	1. Attempt to delete a subject	Any subject	No delete button/option visible; action denied
COURSE & SECTION MANAGEMENT				
TC-R28	Add new section	1. Go to Courses & Sections 2. Click Add Section 3. Save	Section name, course	Section added to list
TC-R29	Edit existing subject	1. Click Edit on subject 2. Update field 3. Save	Any subject	Changes saved

TC #	Test Scenario	Test Steps	Test Data	Expected Result
SETTINGS				
TC-R30	Update security email	1. Settings > Security 2. Change email 3. Save	New valid STI email	Email updated
TC-R31	Change password – valid	1. Change password with valid input	StrongPass@2024	Password saved successfully

Table 6. Test Cases for Registrar

Table 6 above presents the test cases for the Registrar role, which cover schedule and academic records management. Login cases (TC-R01 to TC-R02) confirm that valid credentials grant access to the Registrar dashboard while invalid inputs are denied. SHS Schedule cases (TC-R03 to TC-R11) verify schedule viewing, creation, editing, conflict detection, section locking, publishing, and draft generation, while College Schedule cases (TC-R12 to TC-R14) apply similar operations to tertiary sections, ensuring consistent functionality across academic levels.

Exam Schedule cases (TC-R15 to TC-R18) validate the viewing and management of SHS and College exam schedules, including conflict detection for room assignments. Faculty Management (TC-R19 to TC-R21), Room Management (TC-R22 to TC-R24), Subject Management (TC-R25 to TC-R27), and Course & Section Management (TC-R28 to TC-R29) test the Registrar's ability to view and edit academic records with restricted delete access. Settings cases (TC-R30 to TC-R31) verify account security functions such as updating email and changing passwords. These tests ensure that the Registrar can efficiently manage scheduling and institutional data while preventing conflicts and maintaining data integrity.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-D01	Valid Dept Head login – SHS	1. Enter SHS dept head email & password 2. Login	Shsdepartmenthead @naga.sti.edu Pass@2024	DH Dashboard opens; dept head nav visible
TC-D02	Valid Dept Head login – BSIT	1. Enter BSIT dept head email 2. Login	Bsitdepartmenthead @naga.sti.edu Pass@2024	DH Dashboard for BSIT dept head
TC-D03	Valid Dept Head login – BSHM	1. Enter BSHM dept head email 2. Login	Bshmdepartmenthead @naga.sti.edu Pass@2024	DH Dashboard for BSHM dept head
DH DASHBOARD				
TC-D04	DH Dashboard shows correct dept stats	1. Login as Dept Head 2. View Dashboard	Role: dept_head	Shows faculty count, sections, schedule summary for own department only
CLASS SCHEDULING (DH)				
TC-D05	View class scheduling for own dept	1. Go to Class Scheduling	Role: dept_head	Schedule for own department shown; other depts hidden
TC-D06	View SHS schedule (read access)	1. Go to SHS Schedule	Role: dept_head	Can view SHS schedule; edit access depends on dept
TC-D07	View College schedule (read access)	1. Go to College Schedule	Role: dept_head	Can view college schedule
EXAM SCHEDULING (DH)				
TC-D08	View exam schedule for dept	1. Go to Exam Scheduling	Role: dept_head	Exam entries for own department visible
TC-D09	View SHS/College exam tabs	1. Switch between SHS and College exam tabs	Role: dept_head	Both tabs accessible; entries shown per section
FACULTY MANAGEMENT (DH)				
TC-D10	View faculty within own dept	1. Go to Faculty Management	Role: dept_head	Only faculty from own department listed
TC-D11	Cannot manage faculty of other depts	1. Check faculty list for other departments	Other dept faculty	Other department faculty not shown
SECTION MANAGEMENT (DH)				
TC-D12	View sections for own dept	1. Go to Section Management	Role: dept_head	Sections listed for own department only
TC-D13	Add section to own dept	1. Click Add Section 2. Fill details 3. Save	New section name	Section added under correct dept/strand
TC-D14	Edit section in own dept	1. Click Edit on section 2. Modify 3. Save	Any dept section	Changes saved
TC-D15	Delete section blocked	1. Attempt to delete a section	Any dept section	No delete option visible; action denied
SUBJECT MANAGEMENT (DH)				
TC-D14	View subjects for own dept	1. Go to Subject Management	Role: dept_head	Subjects listed for own dept's courses/strands

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-D15	Add subject to strand/course	1. Click Add Subject 2. Enter name 3. Save	Subject name	Subject added to correct strand/course
REPORTS & EXPORTS (DH)				
TC-D16	Generate reports for own dept	1. Go to Reports & Exports	Role: dept_head	Reports generated for own department; export option available
SETTINGS				
TC-D17	Update security email	1. Settings > Security 2. Change email 3. Save	New valid STI email	Email updated for dept head account
TC-D18	Change password	1. Enter valid new password 2. Save	StrongPass@2024	Password changed successfully

Table 7. Test Cases for Department Head

Table 7 above presents the test cases for the Department Head role, which operates within a department-scoped access level. Login cases (TC-D01 to TC-D03) verify that SHS, BSIT, and BSHM department heads can log in and are directed to their respective dashboards. The Dashboard case (TC-D04) confirms that statistics displayed are limited to the department head's own department. Class Scheduling (TC-D05 to TC-D07) and Exam Scheduling (TC-D08 to TC-D09) cases verify read-only access to schedules within their department only.

Faculty Management (TC-D10 to TC-D11) and Section Management (TC-D12 to TC-D13) cases confirm that only department-specific faculty and sections are accessible. Subject Management (TC-D14 to TC-D15) and Reports (TC-D16) cases test subject viewing and department-scoped export. Settings cases (TC-D17 to TC-D18) validate security email updates and password changes. These tests confirm that the Department Head can effectively manage departmental resources within the boundaries of their assigned role.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-SA01	Valid School Administrator login	1. Enter School Admin email & password 2. Login	schooladmin@naga.sti.edu Pass@2024	Dashboard opens; nav shows
TC-SA02	Invalid credentials	1. Enter wrong email or password	Wrong credentials	Error shown; access denied
DASHBOARD				
TC-SA03	View admin dashboard stats	1. Login as School Admin 2. View Dashboard	Role: School Admin	Dashboard shows schedule summary, faculty count, room availability
SHS SCHEDULE (VIEW)				
TC-SA04	View SHS schedule	1. Go to Senior High School	Role: School Admin	Can view SHS schedule across all sections; read-only
TC-SA05	Cannot add/edit/delete SHS entries	1. Attempt to add or edit SHS entry	Role: School Admin	No Add/Edit/Delete controls visible or accessible
COLLEGE SCHEDULE (VIEW)				
TC-SA06	View College schedule	1. Go to Tertiary	Role: School Admin	Can view college schedule across all sections; read-only
EXAM SCHEDULE (VIEW)				
TC-SA07	View SHS exam schedule	1. Go to Exam > SHS tab	Role: School Admin	All SHS exam entries visible; read-only
TC-SA08	View College exam schedule	1. Go to Exam > College tab	Role: School Admin	All College exam entries visible; read-only
FACULTY (VIEW)				
TC-SA09	View faculty list	1. Go to Faculty	Role: School Admin	All faculty visible with dept and unit info; read-only
ROOM AVAILABILITY (VIEW)				
TC-SA10	View all rooms	1. Go to Room Availability	Role: School Admin	All rooms shown with status; read-only
AUDIT LOGS				
TC-SA11	View audit logs	1. Go to Audit Logs	Role: School Admin	Full audit log accessible; all actions visible
TC-SA12	Cannot access User Management	1. Attempt to navigate to Users page	Role: School Admin	Nav item not visible; access denied
SETTINGS				
TC-SA13	Update security email	1. Settings > Security 2. Update email 3. Save	New valid STI email	Email updated
TC-SA14	Change password	1. Enter valid new password 2. Save	StrongPass@2024	Password changed successfully

Table 8. Test Cases for School Administrator

Table 8 above presents the test cases for the School Administrator role, which is strictly limited to view and oversight access. The login test cases (TC-SA01 to TC-SA02) verify that valid credentials successfully open the school administrator dashboard, while invalid credentials are properly denied access. The Dashboard test case (TC-SA03) confirms that the system displays a read-only overview of schedules, faculty count, and room availability, ensuring that no editing functions are available for this role.

Furthermore, the SHS Schedule (TC-SA04 to TC-SA05), College Schedule (TC-SA06), and Exam Schedule (TC-SA07 to TC-SA08) test cases ensure that all schedules are visible across different sections without any Add, Edit, or Delete controls. The Faculty (TC-SA09) and Room Availability (TC-SA10) cases validate read-only access to institutional data, while Audit Logs (TC-SA11 to TC-SA12) confirm full visibility of system activities without user management privileges. Lastly, the Settings test cases (TC-SA13 to TC-SA14) verify that the role can update security email information and change passwords while maintaining restricted access to other administrative functions.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-F01	Valid Faculty login	1. Enter faculty email & password 2. Login	a.delacruz @naga.sti.edu.ph Pass@2024	Faculty Portal opens; nav shows: Dashboard, SHS Schedule, Tertiary Schedule, Exam & Proctoring, Teaching History, Settings
TC-F02	Faculty cannot access admin pages	1. Login as Faculty 2. Attempt to access User Management or Audit Logs	Role: faculty	Nav items not visible; access denied if URL forced
FACULTY DASHBOARD				
TC-F03	View faculty dashboard	1. Login as Faculty 2. View Dashboard	Role: faculty	Dashboard shows own assigned subjects, total units, upcoming schedule
SHS SCHEDULE (FACULTY VIEW)				
TC-F04	View own SHS schedule	1. Go to SHS Schedule	Role: faculty	Only subjects assigned to this faculty shown; read-only

TC #	Test Scenario	Test Steps	Test Data	Expected Result
TC-F05	Cannot see other faculty schedules	1. View SHS schedule	Other faculty assignments	Other faculty's subjects not displayed
COLLEGE SCHEDULE (FACULTY VIEW)				
TC-F06	View own college schedule	1. Go to Tertiary Schedule	Role: faculty	Only own assigned college subjects shown; read-only
EXAM & PROCTORING				
TC-F07	View own proctoring assignments	1. Go to Exam & Proctoring	Role: faculty	Own proctoring duties shown with room, date, time; read-only
TC-F08	Cannot see other faculty proctoring	1. View exam schedule	Other faculty proctoring	Only own assignments visible
TEACHING HISTORY				
TC-F09	View teaching history	1. Go to Teaching History	Role: faculty	History of subjects taught across previous terms displayed
TC-F10	History filtered by AY/Term	1. Change AY/Term filter in history	Different AY or Term	History updates to show selected period
NOTIFICATIONS				
TC-F11	Faculty receives schedule assignment notification	1. Admin assigns faculty to a class 2. Faculty logs in	Faculty: assigned to new class	Notification visible in bell icon; shows class details
SETTINGS				
TC-F12	Update security email	1. Settings > Security 2. Change email 3. Save	New valid STI email	Email updated for faculty account
TC-F13	Change password – valid	1. Enter valid new password 2. Save	StrongPass@2024	Password changed successfully
TC-F14	Change password – weak password rejected	1. Enter weak password	PW: pass	Password policy error shown; not saved

Table 9. Test Cases for Faculty

Table 9 above presents the test cases for the Faculty role, which is limited to accessing only their own schedule and teaching-related information. Login test cases (TC-F01 to TC-F02) confirm that valid credentials grant access to the Faculty Portal, while attempts to access restricted administrative pages such as User Management and Audit Logs are properly denied. The Dashboard test case (TC-F03) ensures that only the faculty member's assigned subjects, total units, and upcoming schedules are displayed, reinforcing role-based access control.

SHS Schedule (TC-F04 to TC-F05) and College Schedule (TC-F06) test cases validate that faculty members can view only their assigned schedules in a read-only format. Exam & Proctoring (TC-F07 to TC-F08) ensures that only assigned proctoring duties are visible, while Teaching History (TC-F09 to TC-F10) confirms access to past records with filtering by academic year and term. Notifications (TC-F11) verify that faculty receive real-time alerts for new schedule assignments and updates. Lastly, Settings (TC-F12 to TC-F14) test cases ensure that faculty users can update their security email and change passwords in accordance with the system's password policy.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
LOGIN				
TC-S01	Valid Student login	1. Enter student email & password 2. Login	j.cruz@naga.sti.edu.ph Pass@2024	Student Dashboard opens
TC-S02	Student cannot access admin/faculty pages	1. Login as Student 2. Attempt to access Faculty or User Management	Role: student	Nav items not visible; access denied
STUDENT DASHBOARD				
TC-S03	View student dashboard	1. Login as Student 2. View Dashboard	Role: student	Dashboard shows own section info, upcoming classes, announcements
CLASS SCHEDULE (STUDENT)				
TC-S04	View own class schedule	1. Go to Class Schedule	Role: student	Only schedule for student's own section shown
TC-S05	Cannot see other sections' schedules	1. View class schedule	Other section data	Only own section schedule visible; no section switching
TC-S06	Schedule shows correct subject, faculty, room, day, time	1. View class schedule	Student section: ABM-11A	All schedule details correctly displayed per subject
TC-S07	Published schedules visible; unpublished hidden	1. View class schedule	Section with mixed published/unpublished	Only published entries visible to student

TC #	Test Scenario	Test Steps	Test Data	Expected Result
EXAM SCHEDULE (STUDENT)				
TC-S08	View own exam schedule	1. Go to Exam Schedule	Role: student	Exam entries for own section shown with subject, room, date, time
TC-S09	Cannot see other sections' exams	1. View exam schedule	Other section exam data	Only own section exams visible
TC-S10	Exam schedule shows grading Period	1. View exam schedule	Midterm/Final grading period	Correct grading period label shown (e.g. Midterm, Final)
NOTIFICATIONS				
TC-S11	Faculty receives schedule assignment notification	1. Admin assigns faculty to a class 2. Faculty logs in	Faculty: assigned to new class	Notification visible in bell icon; shows class details
SETTINGS				
TC-S12	Update security email	1. Settings > Security 2. Change email 3. Save	New valid STI email	Email updated for student account
TC-S13	Change password – valid	1. Enter strong valid password 2. Save	StrongPass@2024	Password changed successfully
TC-S14	Change password – weak rejected	1. Enter weak password 2. Save	PW: abc	Password policy error; not saved

Table 10. Test Cases for Student

Table 10 presents the Student role test cases, limited to viewing their own class and exam schedules. Login tests (TC-S01–TC-S02) validate access to the student dashboard while restricting faculty and administrative pages. The Dashboard (TC-S03) ensures correct display of section details, upcoming classes, and announcements.

Class Schedule (TC-S04–TC-S07) and Exam Schedule (TC-S08–TC-S10) verify that students can only view their own schedules, with unpublished entries hidden and details accurately displayed. Notifications (TC-S11) confirm real-time alerts for updates, while Settings (TC-S12–TC-S14) ensure students can update their security email and change passwords in accordance with system policy.

TC #	Test Scenario	Test Steps	Test Data	Expected Result
RESPONSIVE DESIGN				
TC-RD01	Mobile layout – login page	1. Open STIzen on a mobile browser 2. View login page	Device: Mobile (375px width)	Login form is fully visible, properly scaled, and usable without horizontal scrolling
TC-RD02	Mobile layout – dashboard	1. Login on mobile browser 2. View dashboard	Device: Mobile (375px width)	Dashboard widgets and statistics stack vertically and are fully readable
TC-RD03	Mobile layout – schedule table	1. Login on mobile 2. Navigate to Class Schedule	Device: Mobile (375px width)	Schedule table is scrollable or reformatted; no content is cut off
TC-RD04	Tablet layout – navigation	1. Open STIzen on a tablet browser 2. Check sidebar navigation	Device: Tablet (768px width)	Sidebar collapses or remains accessible and properly displayed
TC-RD05	Desktop layout – full view	1. Open STIzen on desktop browser 2. Navigate all modules	Device: Desktop (1280px width)	All modules display correctly with full sidebar, tables, and content visible

Table 11. Test Cases for Responsive Design

Table 11 above presents the test cases for Responsive Design, which verify that the STIzen system is fully accessible and functional across different screen sizes and devices. TC-RD01 to TC-RD03 confirm that the login page, dashboard, and schedule tables render and behave correctly on mobile screens (375px), ensuring no content is cut off and all elements are readable. These tests also validate that interactive components such as buttons, menus, and tables remain usable and properly aligned on smaller screens.

TC-RD04 verifies proper navigation display on tablet screens (768px), while TC-RD05 ensures the complete layout is intact on desktop screens (1280px). These tests collectively confirm that all user roles can interact with the system comfortably regardless of the device they use, with browser developer tools such as Chrome DevTools used to simulate and verify responsive behavior across all screen sizes. The results ensure consistent user experience and interface stability across varying screen resolutions and device types.

Gantt Chart of Activities

The Gantt Chart of Activities serves as a project management tool that outlines the planned workflow and timeline of tasks. It helps track progress and ensures that each phase of the project is completed within the expected timeframe. The Gantt Chart of Activities shows the timeline of tasks for the STIzen system, including their schedule, duration, and order of completion. For this Capstone 1, only the first two phases of the project are presented, focusing on the initial planning and development activities.

Activity	February 2026				March 2026				April 2026				May 2026				June 2026			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
PHASE 1 — REQUIREMENTS GATHERING & ANALYSIS																				
Topic identification & proposal																				
Stakeholder interviews (admin, faculty)																				
Survey distribution to students																				
Document review & observation																				
Problem analysis & research questions																				
Literature review (local & foreign)																				
Chapter 1 write-up																				
Chapter 2 write-up																				
PHASE 2 — SYSTEM DESIGN																				
Conceptual framework (IPO model)																				
ERD design																				
DFD (Level 0, 1, 2)																				
Process flow diagrams (all roles)																				
Use case diagrams																				
UI/UX wireframes & prototypes																				
Chapter 3 write-up																				
Key Dates									Apr 7 Mock Defense	April 9 Revisions	April 24 Revisions	Apr 28 Proposal Defense								

Table 12. STIzen Development Timeline Gantt Chart

Table 12 above presents the Gantt chart outlining the project timeline of STIzen from February to June 2026. The development process is divided into two main phases to ensure a structured and organized workflow throughout the project.

To begin with, Phase 1 (Requirements Gathering & Analysis) runs from February to early April 2026. This phase includes topic identification and proposal writing, stakeholder interviews with administrators and faculty, survey distribution to students, document review and observation, and problem analysis leading to the formulation of research questions. It also involves a literature review of both local and foreign studies, as well as the completion of Chapters 1 and 2 to establish the foundation of the system. This phase ensures that all user needs and system requirements are properly identified and analyzed before development begins.

Subsequently, Phase 2 (System Design) begins in March and continues through April 2026, focusing on the design and modeling of the system. Activities include developing the conceptual framework using the IPO model, system architecture design, ERD, DFDs (Levels 0, 1, and 2), process flow diagrams for all user roles, and UI/UX wireframes and prototypes, along with the writing of Chapter 3. Key milestones are also indicated, including the Mock Defense on April 7, revisions on April 9 & 24, and the Proposal Defense on April 28, 2026. Activities scheduled after April 28, 2026, will be dedicated to revisions and polishing of the research paper and system prototype in preparation for the final defense.

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